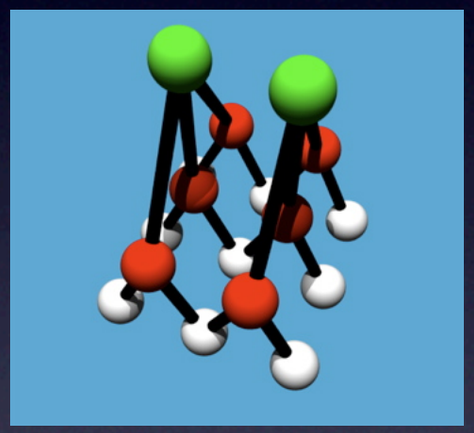
Crazy Quantum Concatenated Code Hamiltonians*



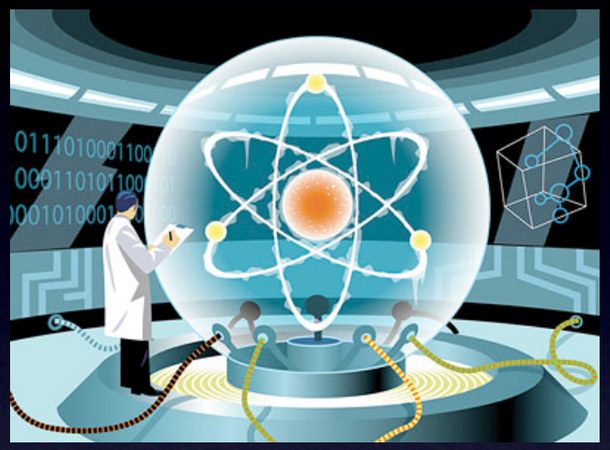
Dave Bacon
University of Washington

Funded by

* This proposal, like all proposals for quantum computation, relies on speculative technology, does not in its current form take into account all possible sources of noise, unreliability and manufacturing error, and probably will not work.

The Quantum Computing Enemy

- Decoherence
- Imprecise control



The Quantum Computing Enemy

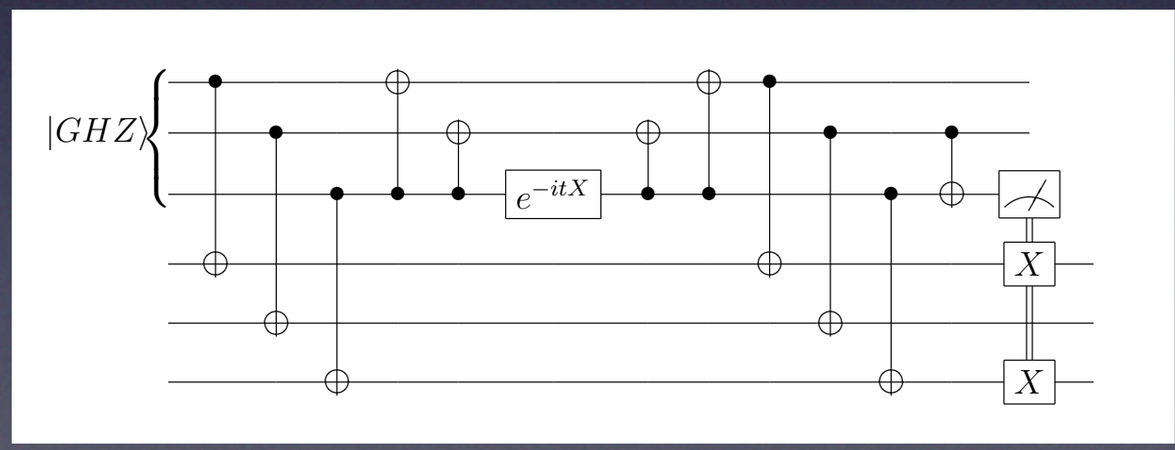
- Decoherence
- Imprecise control

The Quantum Computing Enemy

- Decoherence
- Imprecise control

The Solution?

- Fault-tolerant quantum computing



fight **physics** with **engineering**

Fight Physics with Physics



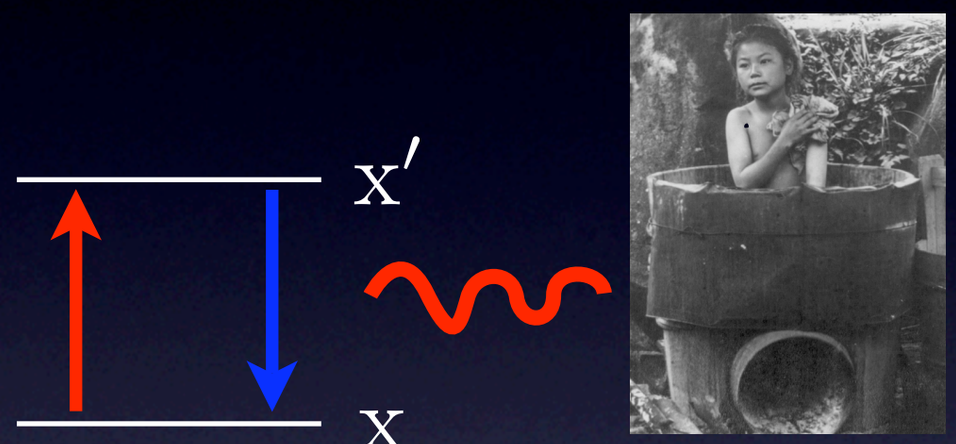
The Computing Phase



Fundamental Question

What makes a physical system a good storer and manipulator of (quantum/classical) information?

Heating and Cooling



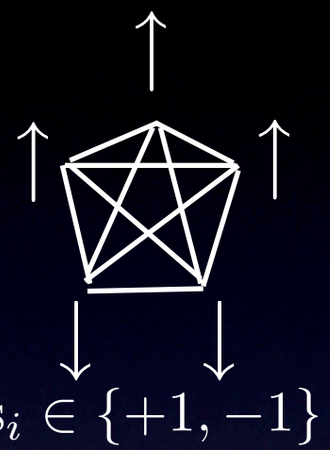
system bath

$$T = \beta^{-1}$$

For perturbative interactions with bath in equilibrium we expect heating to be slower than cooling:

$$\frac{Pr[x \to x']}{Pr[x' \to x]} = e^{-\beta(E[x']-E[x])}$$

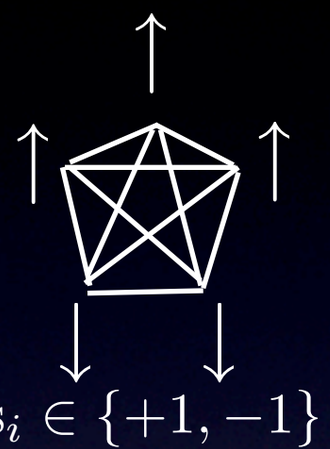
Model Hard Drive



$$E = -J\left(\sum_i s_i\right)^2 = -JNx^2$$

$$x = \frac{1}{N}\sum_i s_i$$

Model Hard Drive



$$E = -J\left(\sum_i s_i\right)^2 = -JNx^2 \qquad x = \frac{1}{N}\sum_i s_i$$

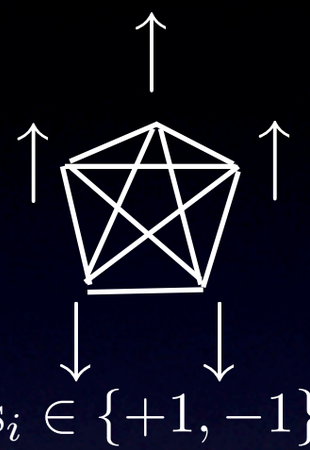
spin flip: $x \to x \pm \frac{2}{N}$

$$\frac{Pr[x \to x']}{Pr[x' \to x]} = e^{-\beta(F[x']-F[x])}$$

$$F = -JNx^2 - NkTh\left(\frac{x+1}{2}\right)$$

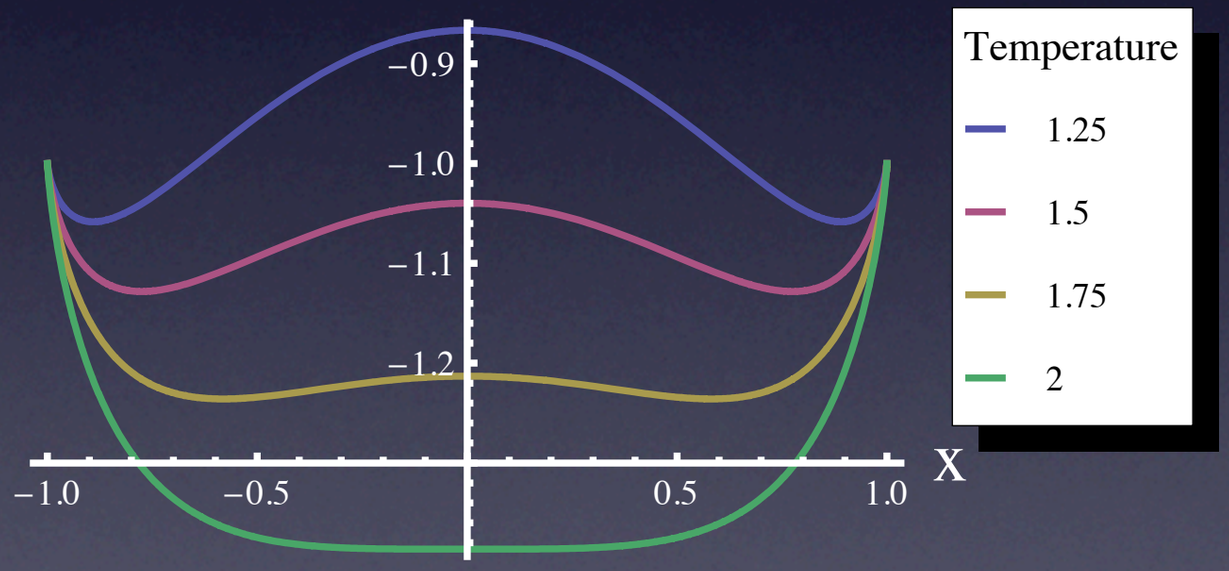
$$h(x) = -x\ln x - (1-x)\ln(1-x)$$

Model Hard Drive



$$E = -J\left(\sum_i s_i\right)^2 = -JNx^2 \qquad x = \frac{1}{N}\sum_i s_i$$

Free Energy



spin flip: $x \to x \pm \frac{2}{N}$

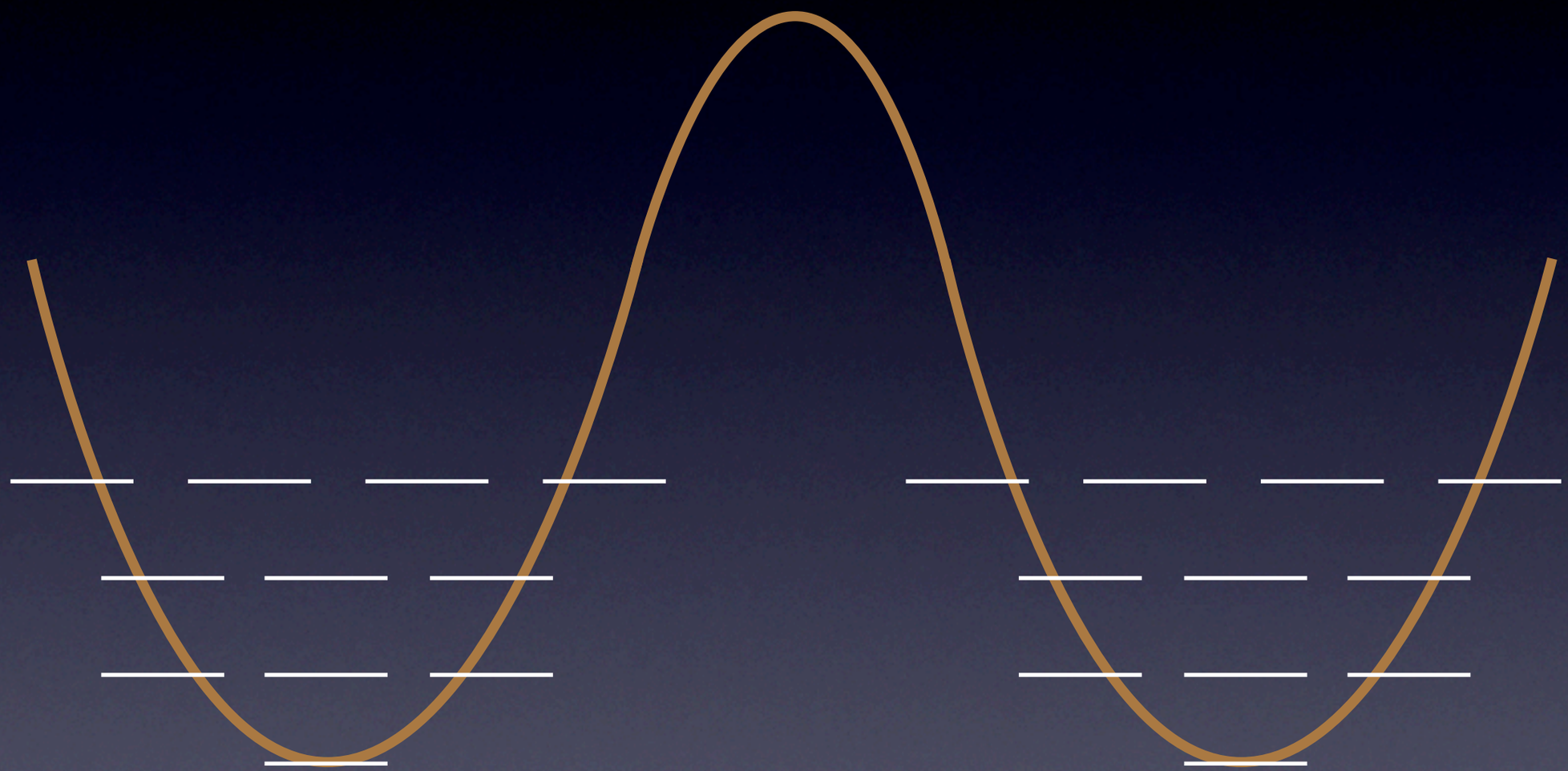
$$\frac{Pr[x \to x']}{Pr[x' \to x]} = e^{-\beta(F[x']-F[x])}$$

“1” =↓↓↓↓↓ “0” =↑↑↑↑↑

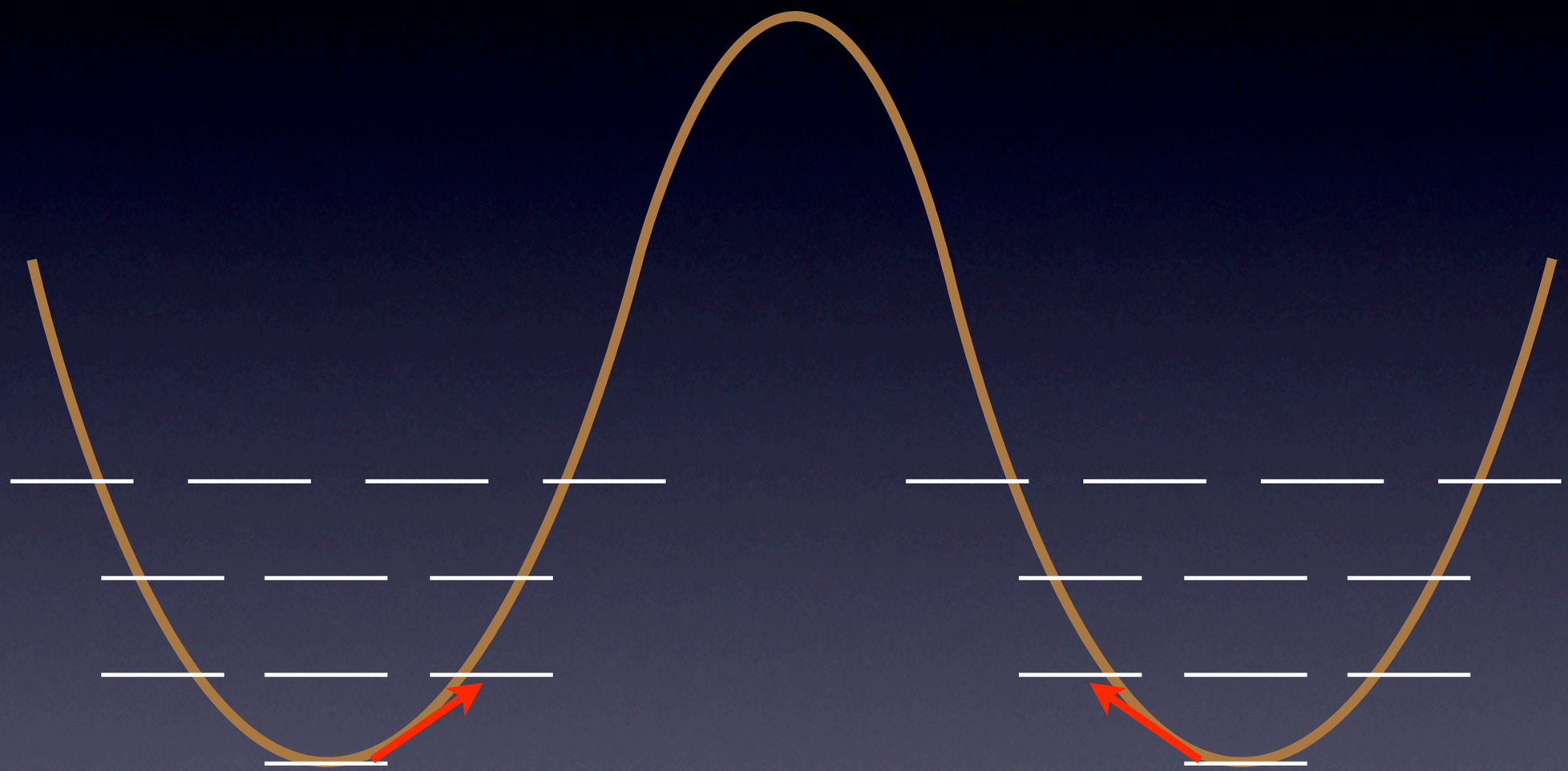
$$F = -JNx^2 - NkTh\left(\frac{x+1}{2}\right)$$

$$h(x) = -x\ln x - (1-x)\ln(1-x)$$

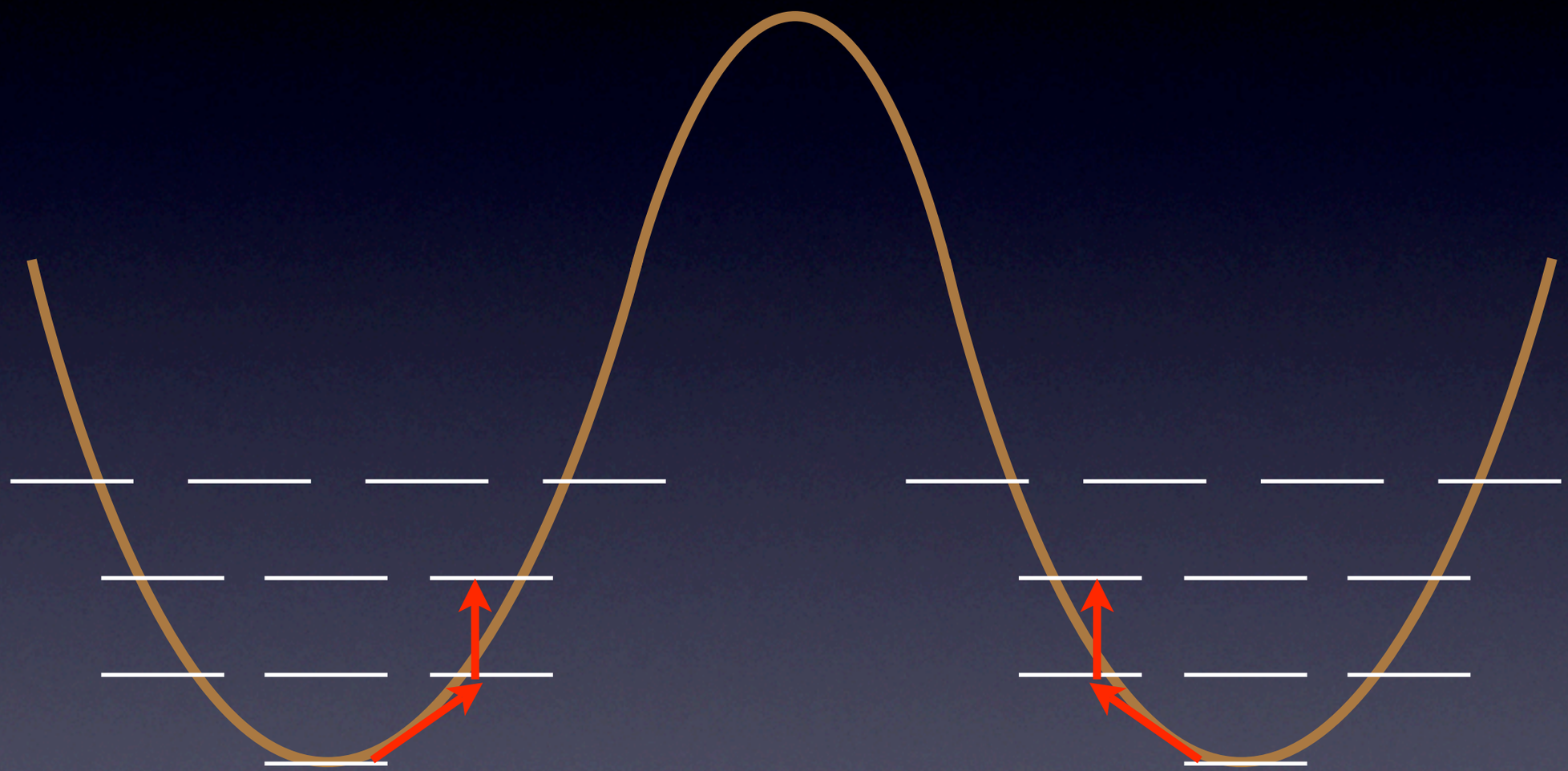
Self-Correction



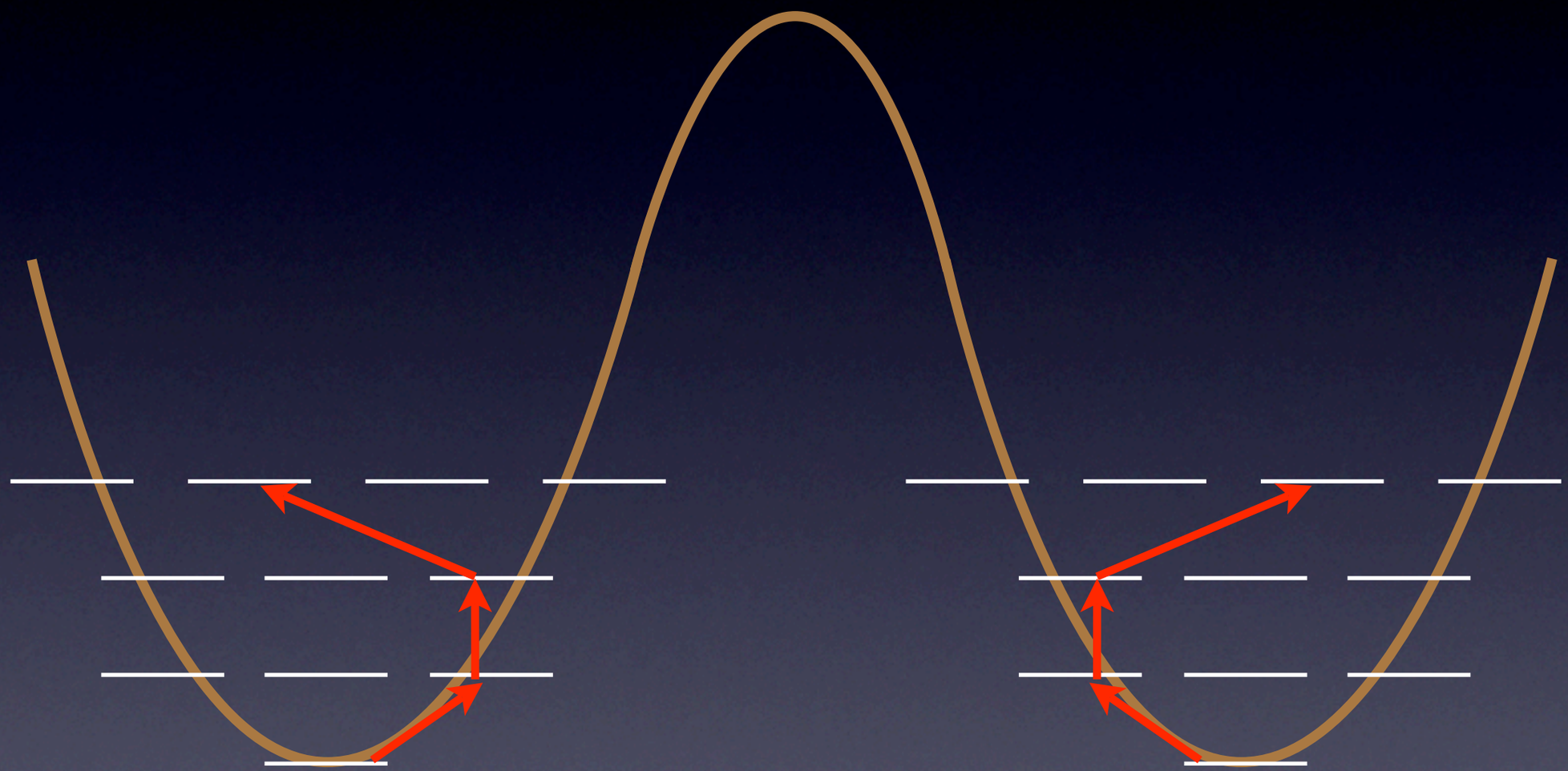
Self-Correction



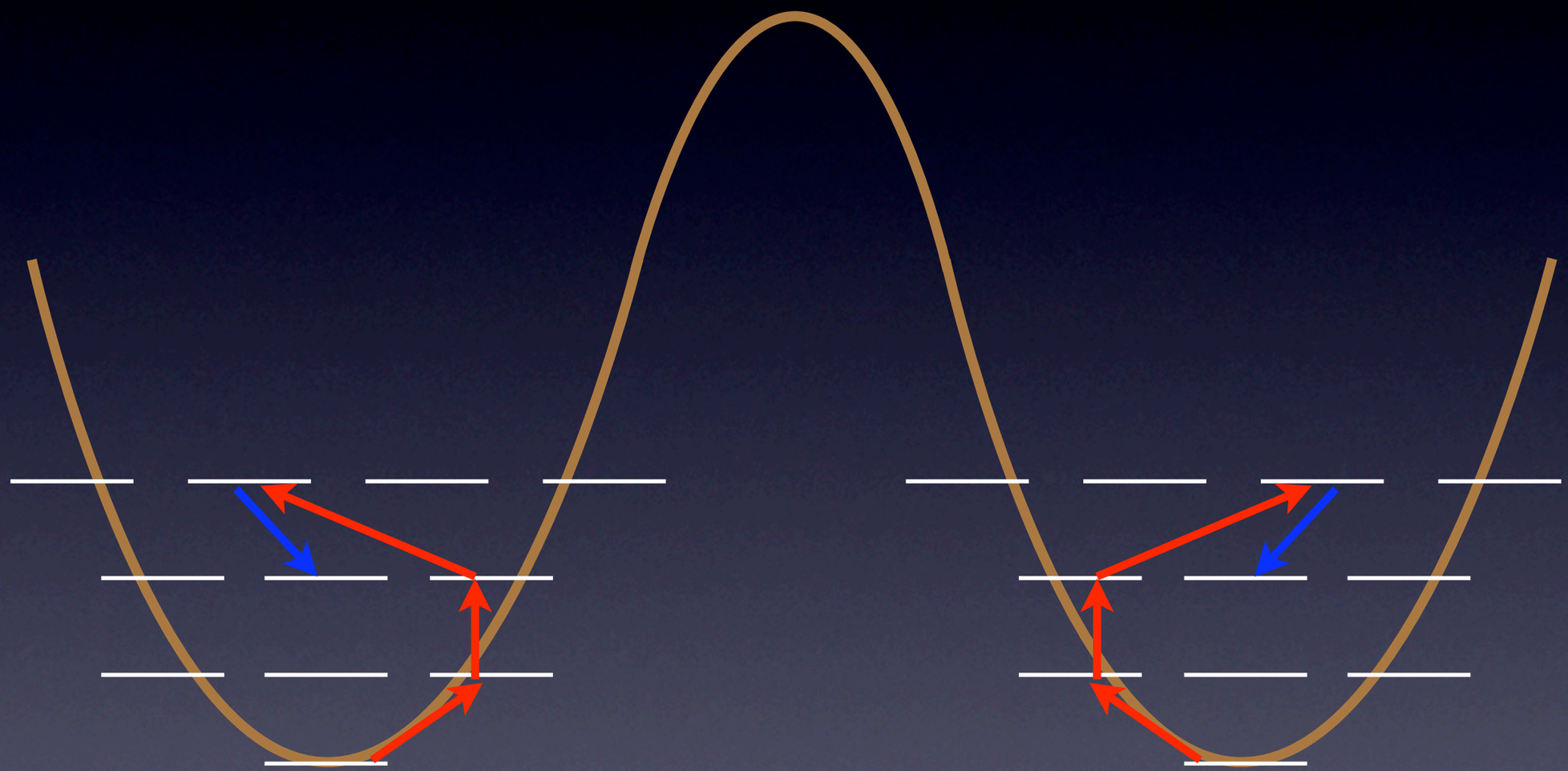
Self-Correction



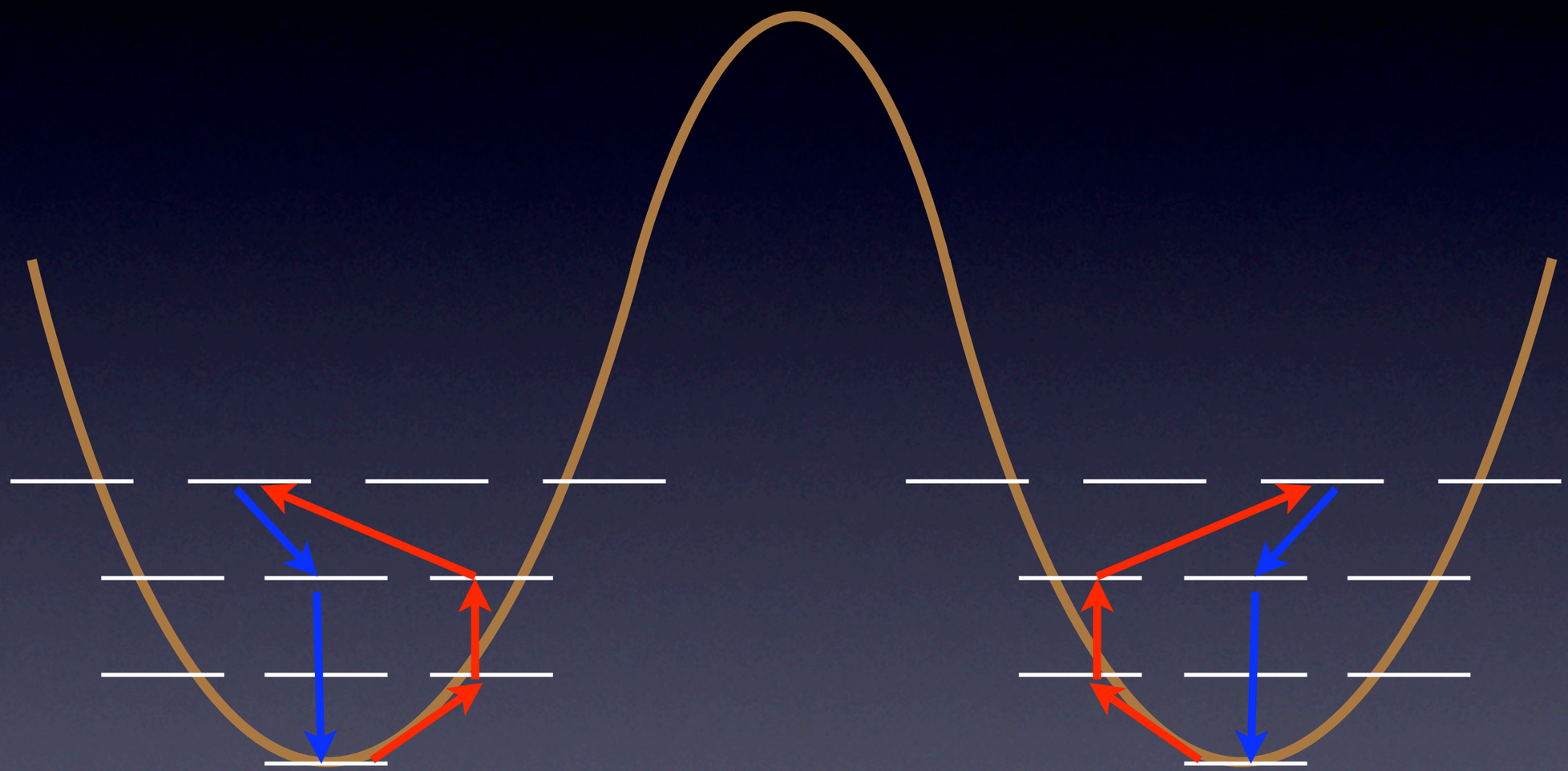
Self-Correction



Self-Correction

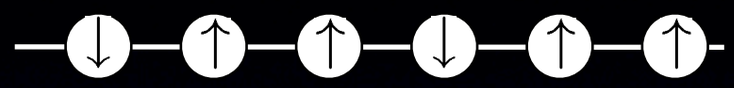


Self-Correction

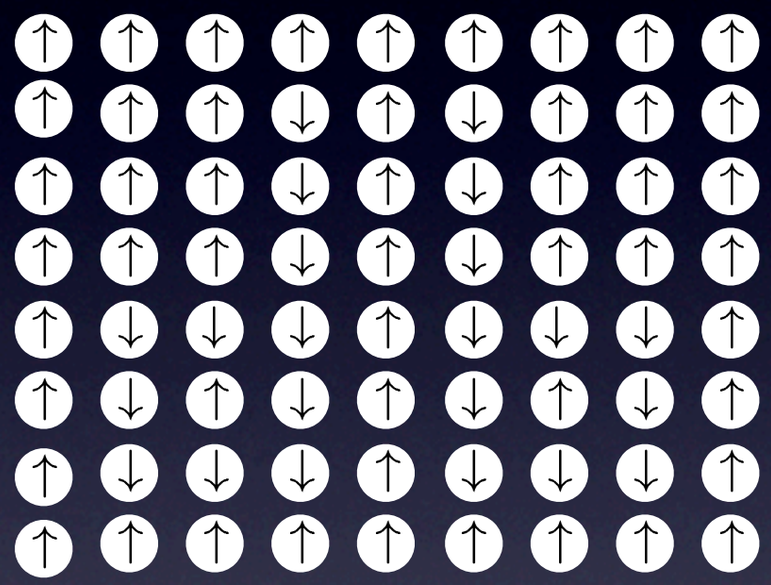


Storage Comparisons

Ferromagnetic Ising models:



one dimensional



two (and >) dimensional

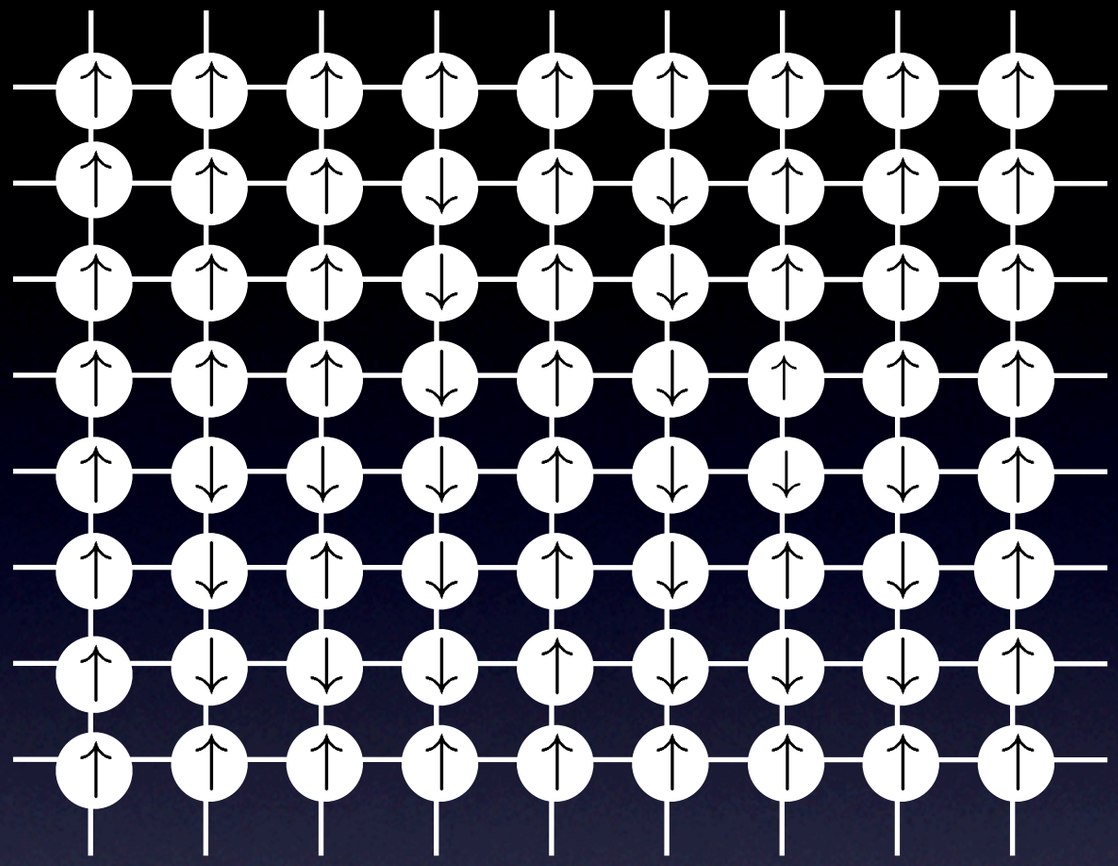
$$E = -J \sum_{\langle i,j \rangle} s_i s_j$$

Toom's rule:

0 0 I 0 0 0 0 0
0 I I 0 I 0 0 0
0 I 0 0 0 0 0 0
0 0 I I I I I 0
0 0 0 0 I 0 I 0
I I I 0 0 0 0 0

two dimensional noisy cellular automata

Dynamic Ising Models





Interaction with environment causes spin flips

Update rule:

- If flipping spin would decrease energy, flip spin
- If flipping spin would increase energy, flip with probability $e^{-\beta \Delta E}$

Toom's Rule



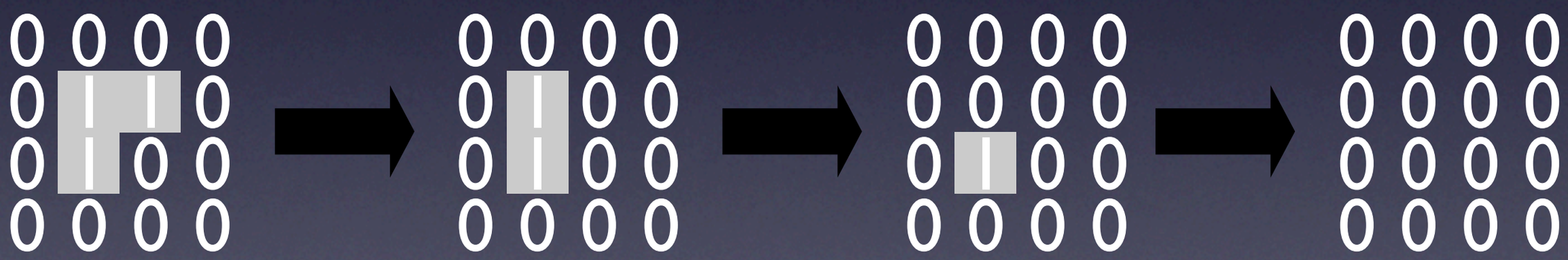
0 0 I 0 0 0 0 0
0 I I 0 I 0 0 0
0 I 0 0 0 0 0 0
0 0 I I I I I 0
0 0 0 0 I 0 I 0
I I I 0 0 0 0 0

two dimensional noisy cellular automata

Update rule:



b=majority(n,e,b)



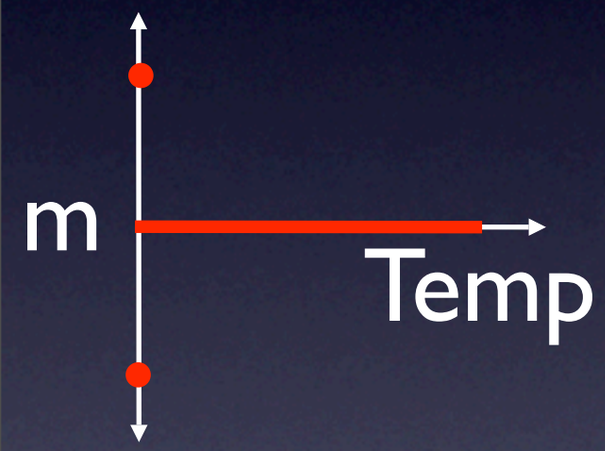
Update rule with noise: with some probability the update fails and a random state is assigned

Storage

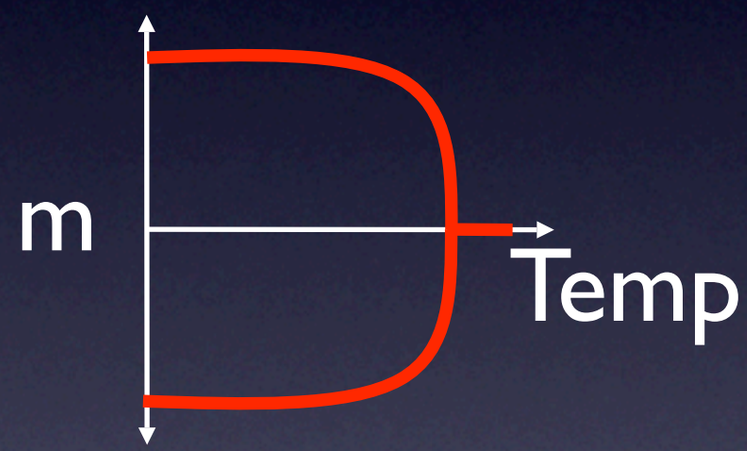
Long time behavior (not infinite time, but effectively)

$$m = \frac{1}{N} \sum_i s_i$$

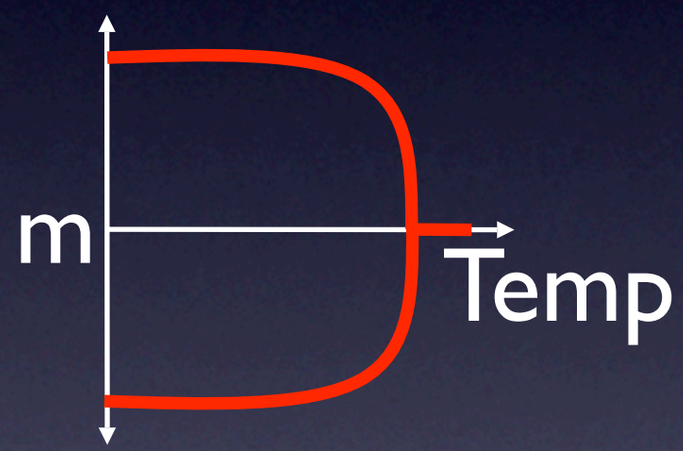
1D Ising



2D Ising



Toom's Rule



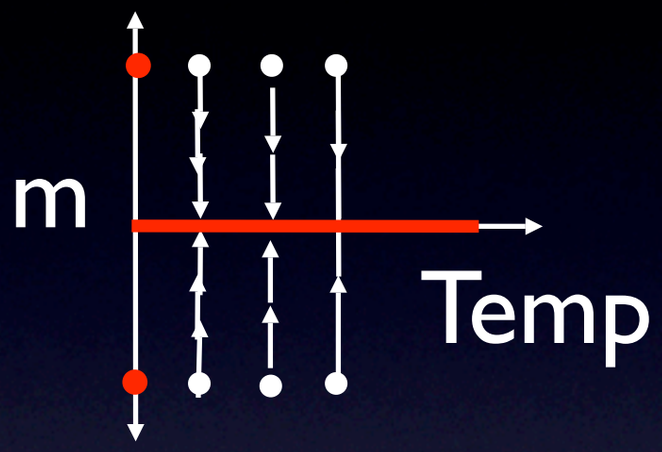
Criticism: (1) thermodynamic limit taken
(2) what if relaxation to equilibrium takes a long time?

Storage



2nd attempt:

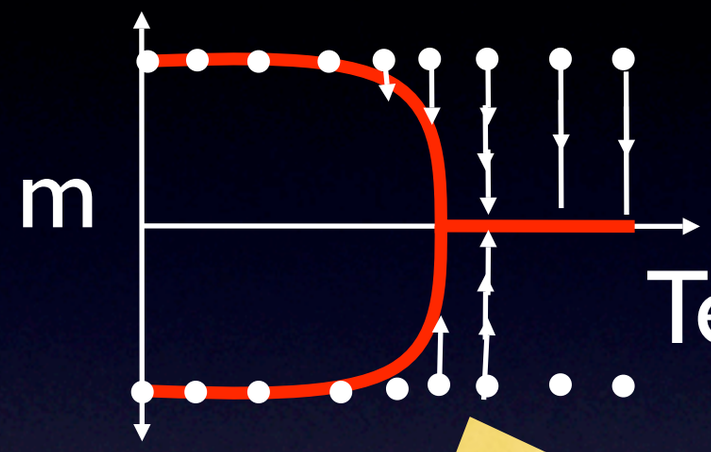
1D Ising



decay rate $\propto e^{-\beta J}$

Relaxation to thermal

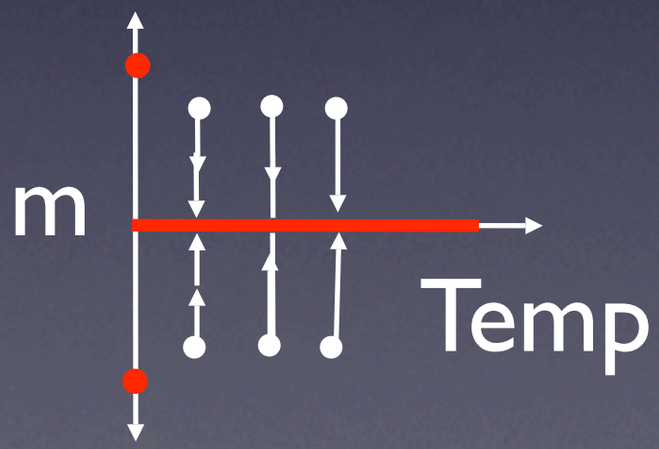
2D Ising, Toom's Rule

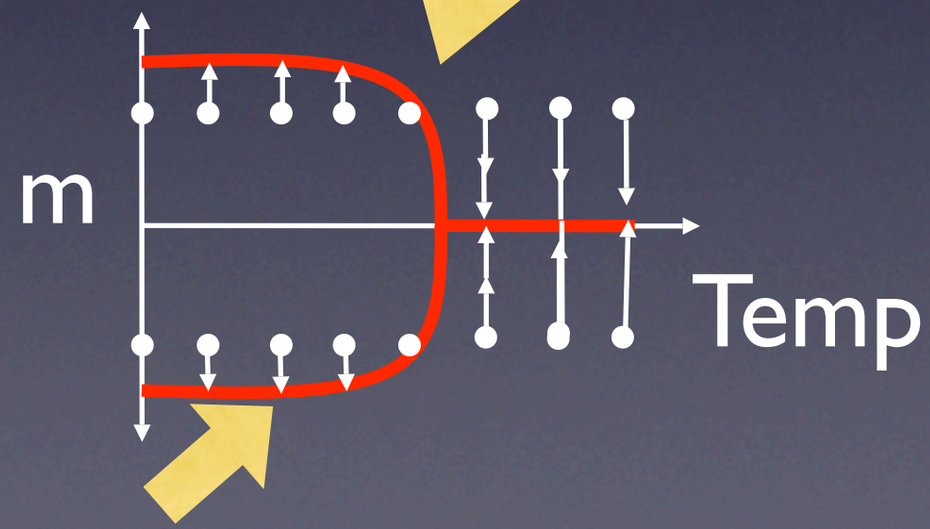


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





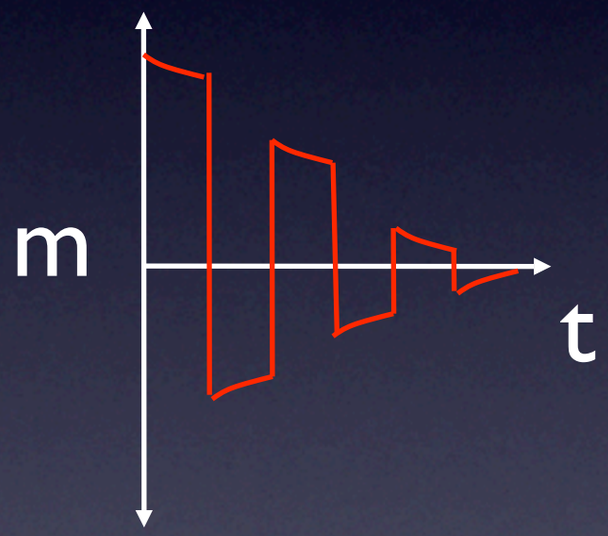
fast relaxation or equilibrium state

Manipulation



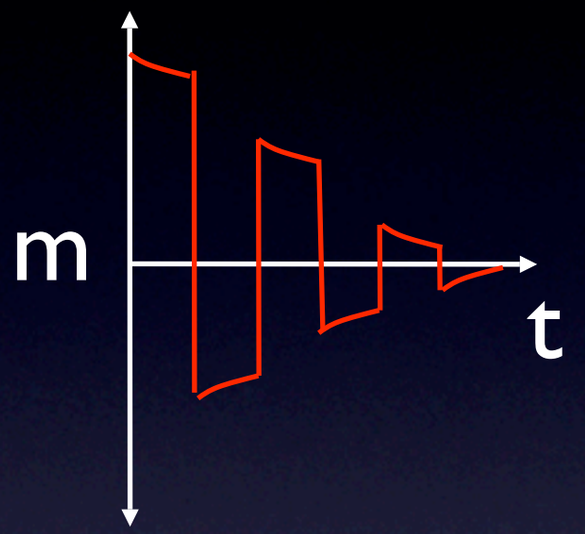
Flipping spins is imperfect

1D Ising

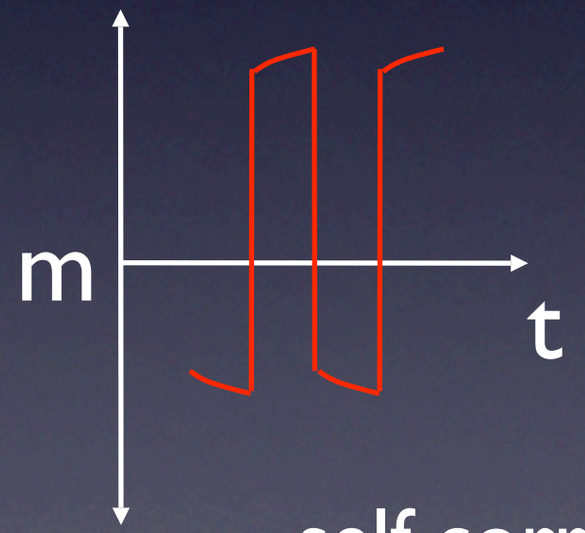


vs

2D Ising
Toom's Rule



$T > T_C$



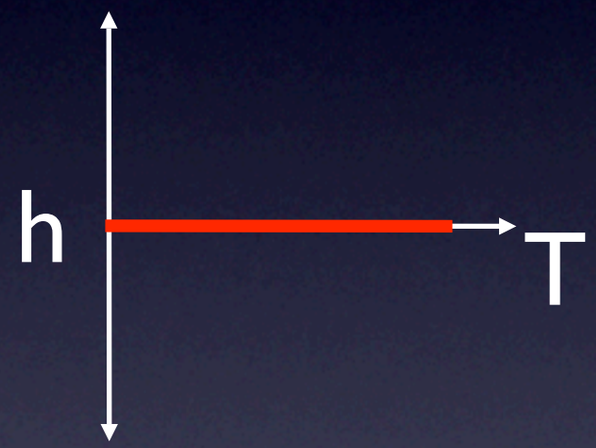
$T < T_C$

self correcting

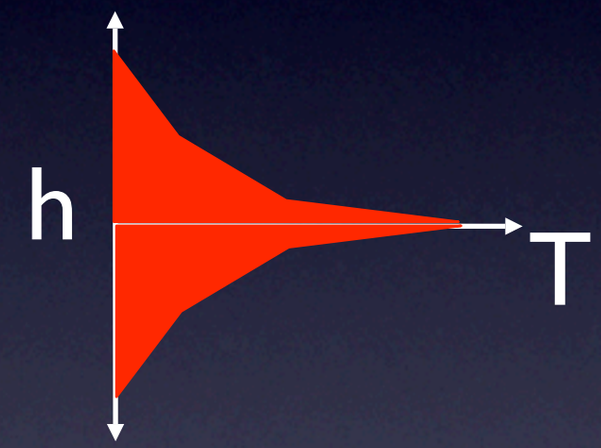
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



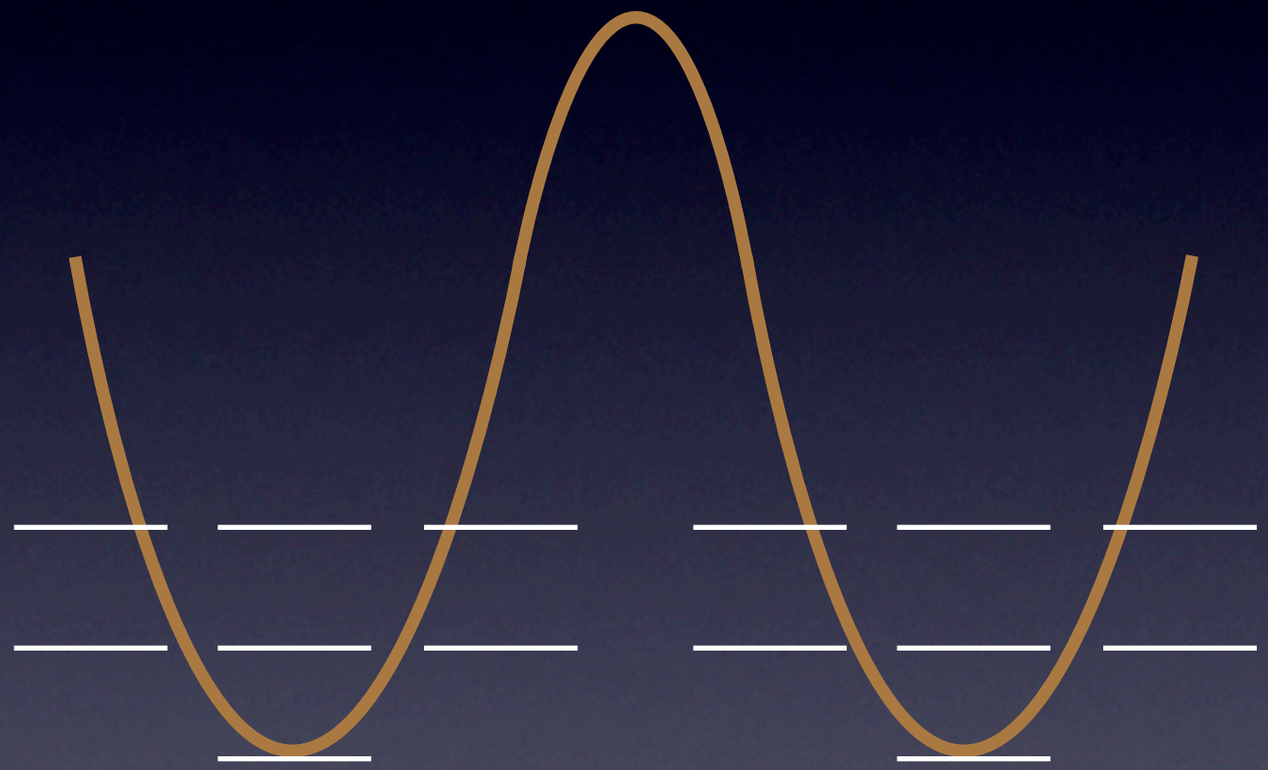
Toom's Rule



Regions with two coexisting phases in red

External magnetic fields eventually disorder magnetic media

Quantum Hard Drives?

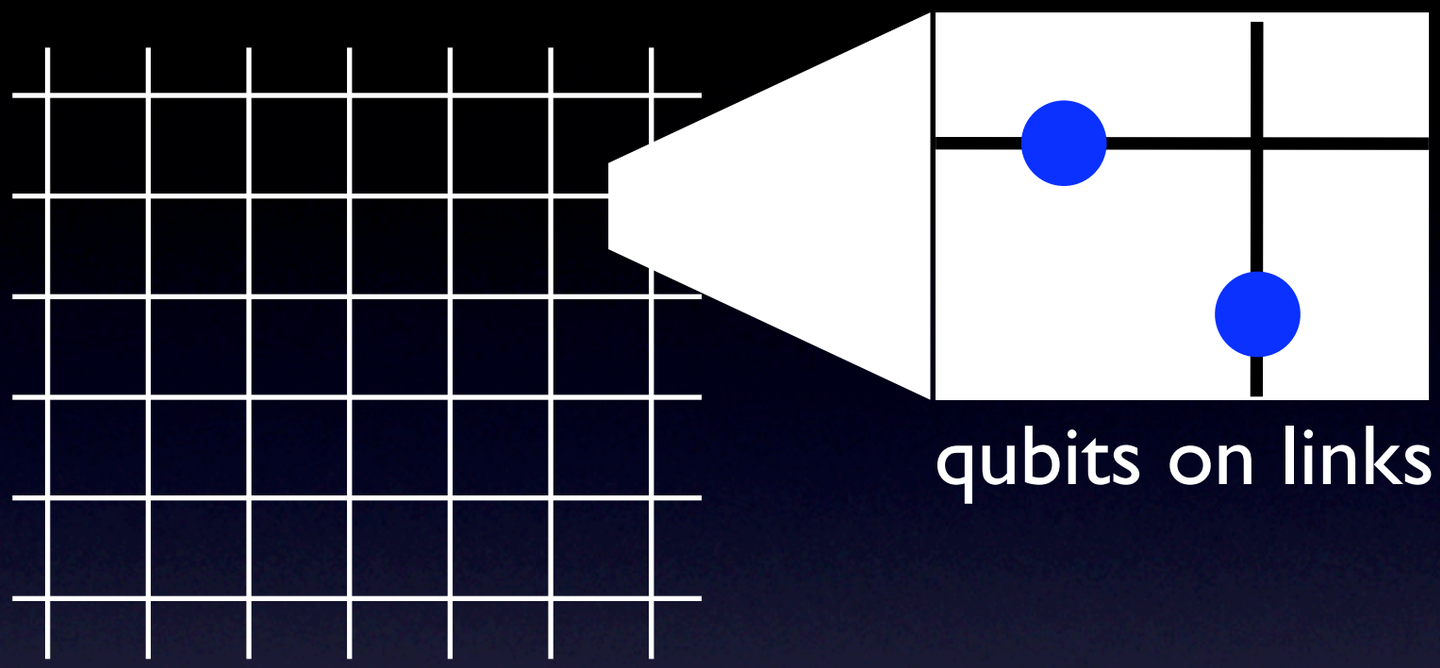


quantum error correcting
code states

Kitaev's Toric Code Model



Kitaev

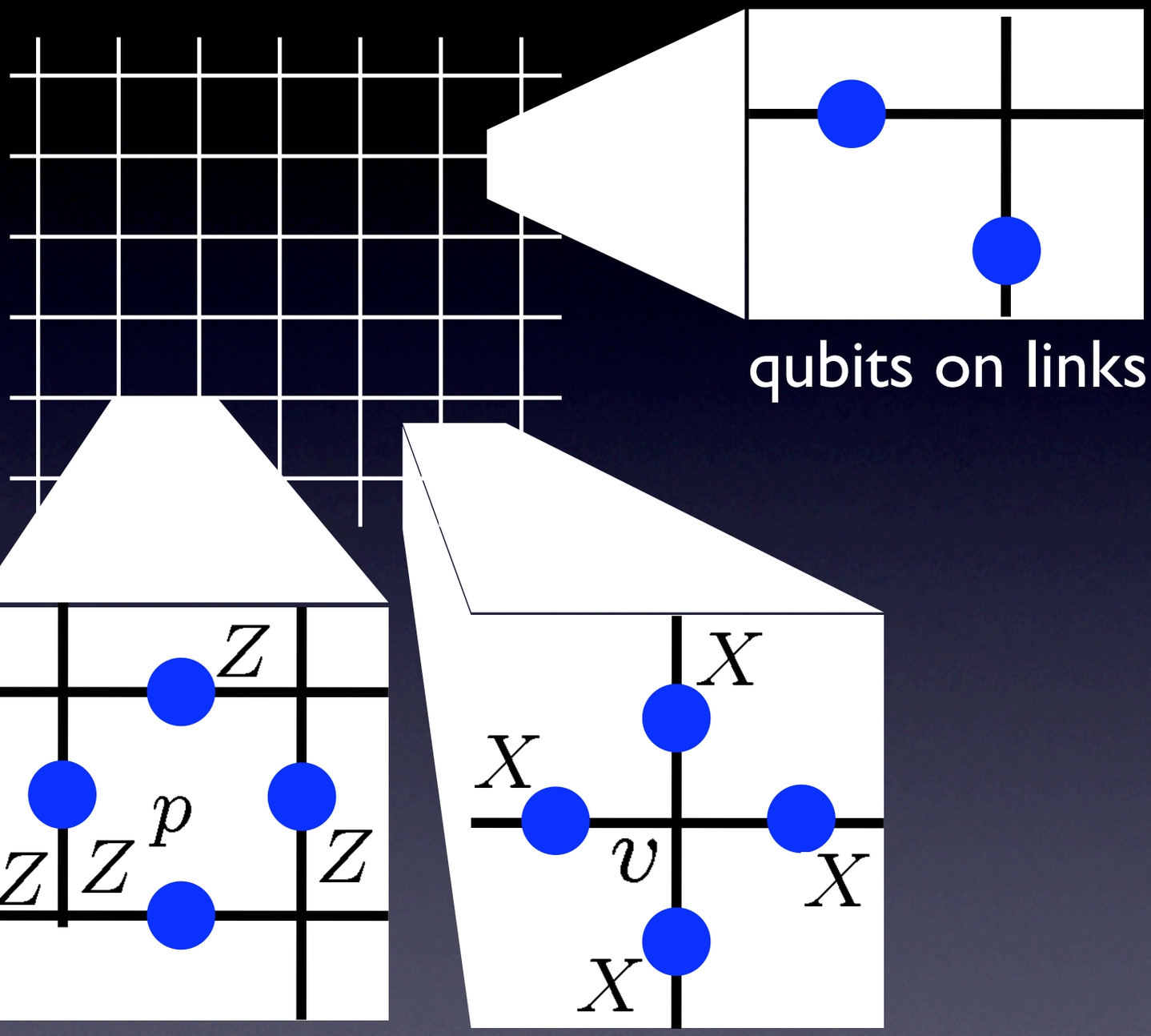


qubits on links

Kitaev's Toric Code Model



Kitaev



plaquette operators:

$$S_p = \prod_{e \in p} Z_e$$

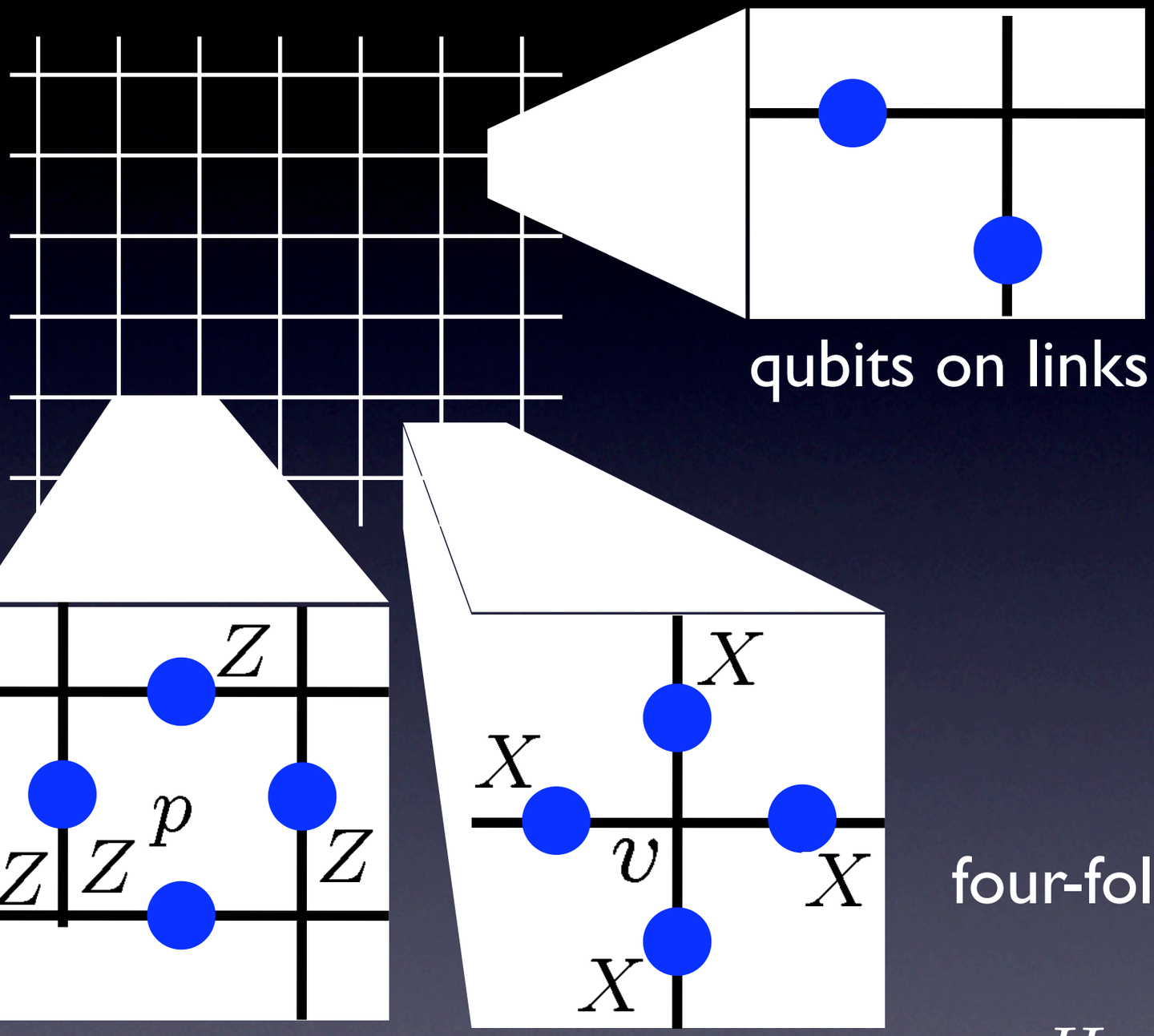
vertex operators:

$$S_v = \prod_{e \in v} X_e$$

Kitaev's Toric Code Model



Kitaev



plaquette operators:

$$S_p = \prod_{e \in p} Z_e$$

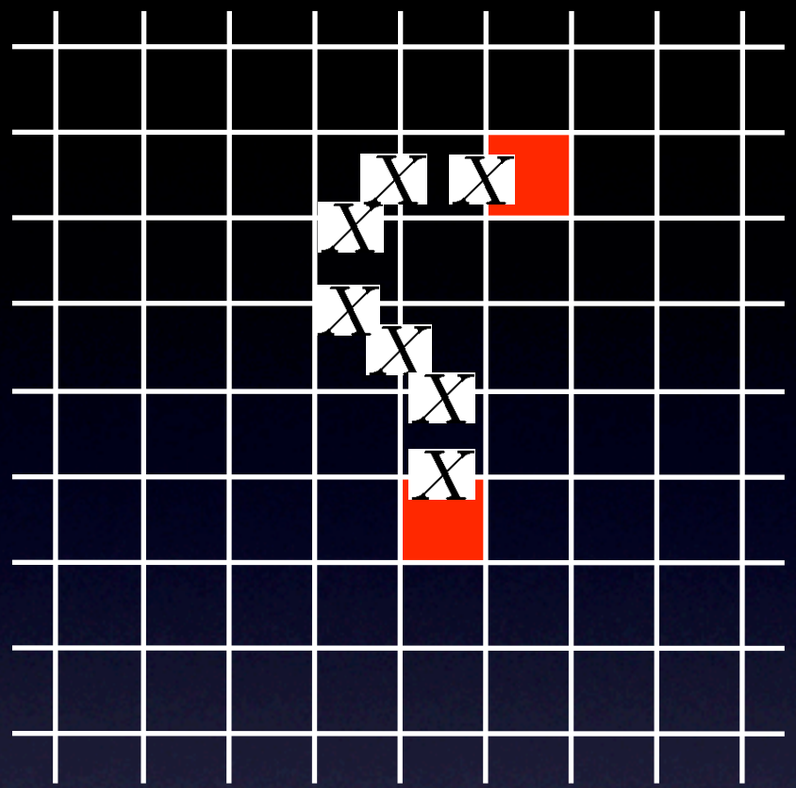
vertex operators:

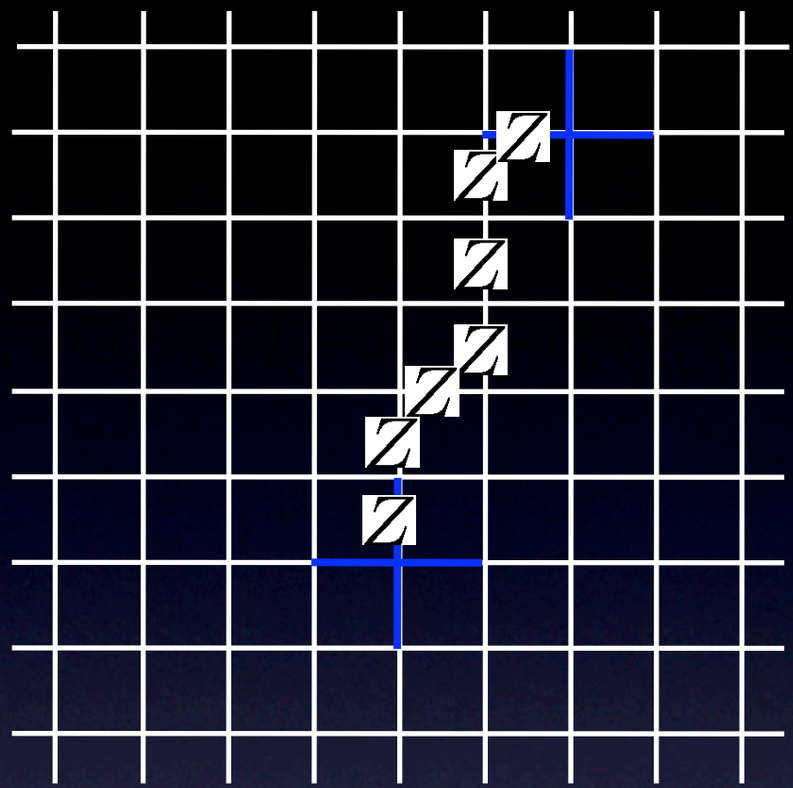
$$S_v = \prod_{e \in v} X_e$$

four-fold degenerate ground state

$$H = -J \left[\sum_v S_v + \sum_p S_p \right]$$

Excitations





excitations particles come in pairs (particle/antiparticle)
at end of “error” chains

two types of particles, X-type (live on vertices of dual lattice)
Z-type (live on vertices of the lattice)



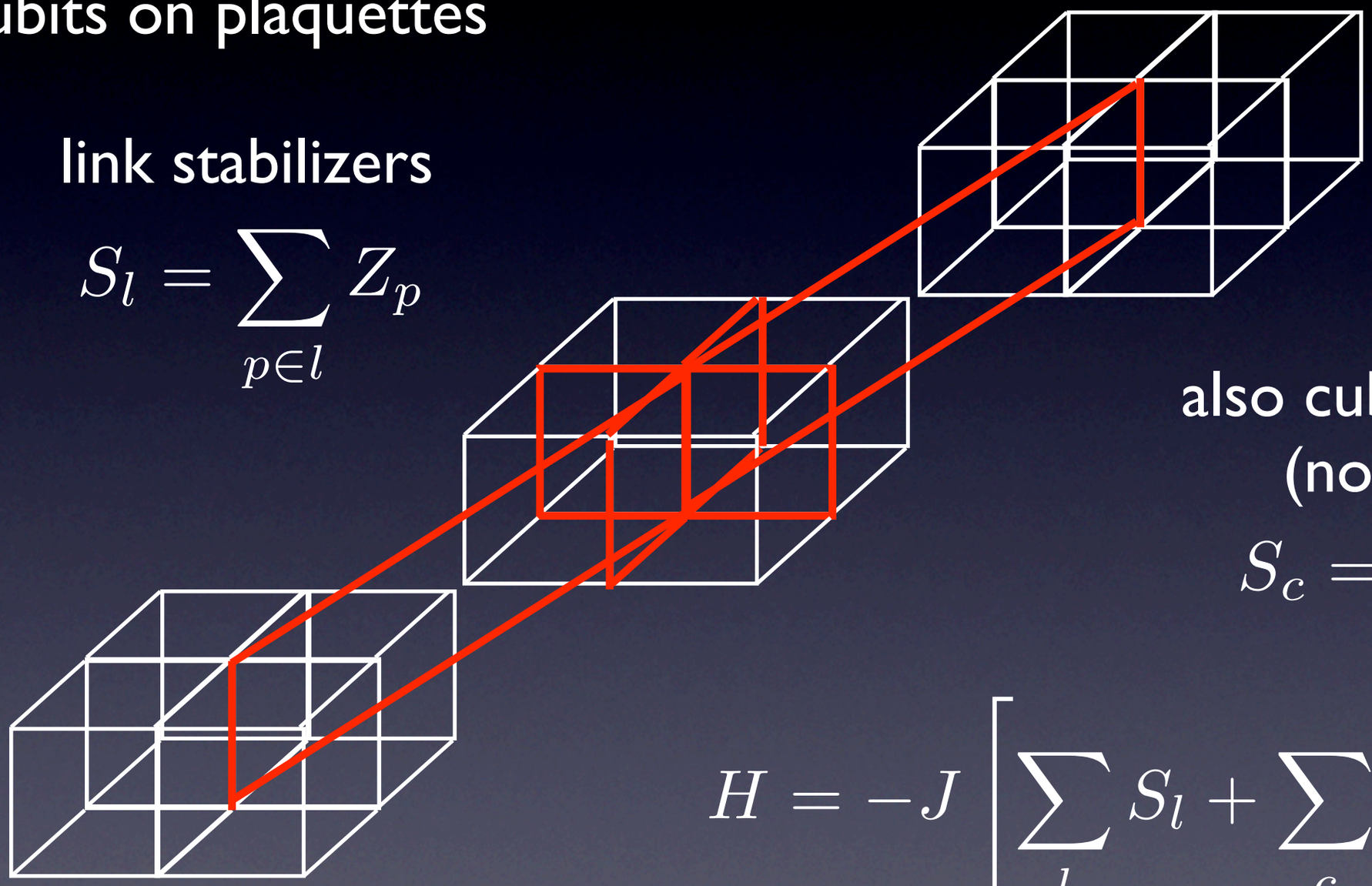
not self correcting...

Into the Fourth Dimesnion

qubits on plaquettes

link stabilizers

$$S_l = \sum_{p \in l} Z_p$$



also cube stabilizers
(not shown)

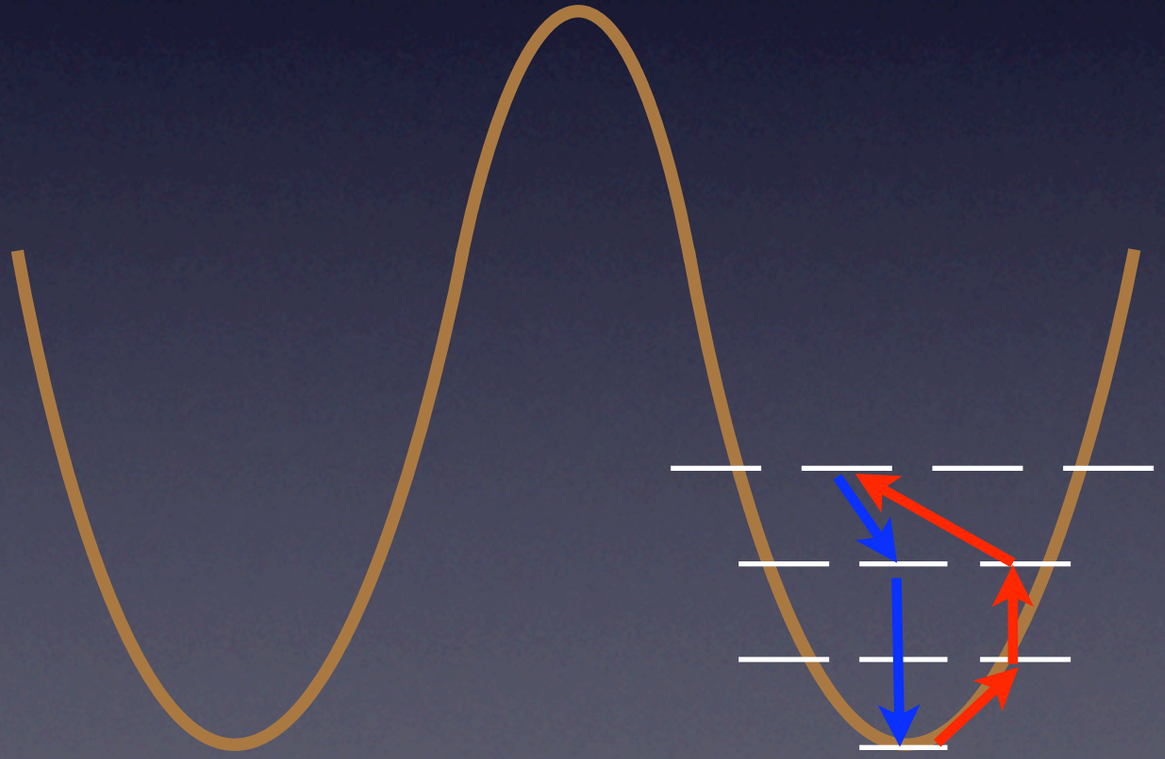
$$S_c = \sum_{p \in c} X_c$$

$$H = -J \left[\sum_l S_l + \sum_c S_c \right]$$

Excitations are Strings



Energy cost proportional to perimeter



$Jn^{1/4}$

self correcting

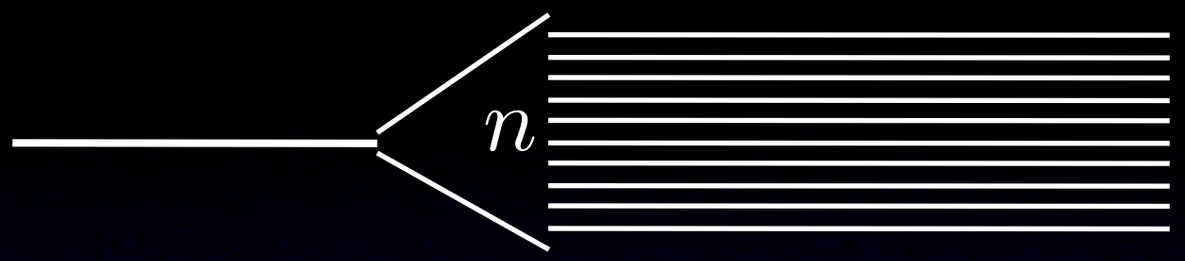
[Dennis, Landahl, Preskill, Kitaev 2002]

Perfection Through Concatenation

failure probability

$$\frac{}{p}$$

Perfection Through Concatenation

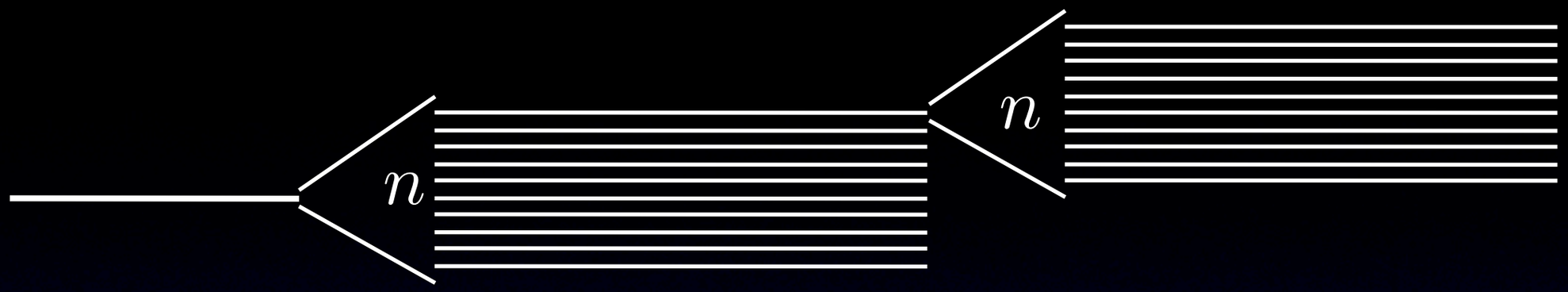


failure probability

p

cp^2

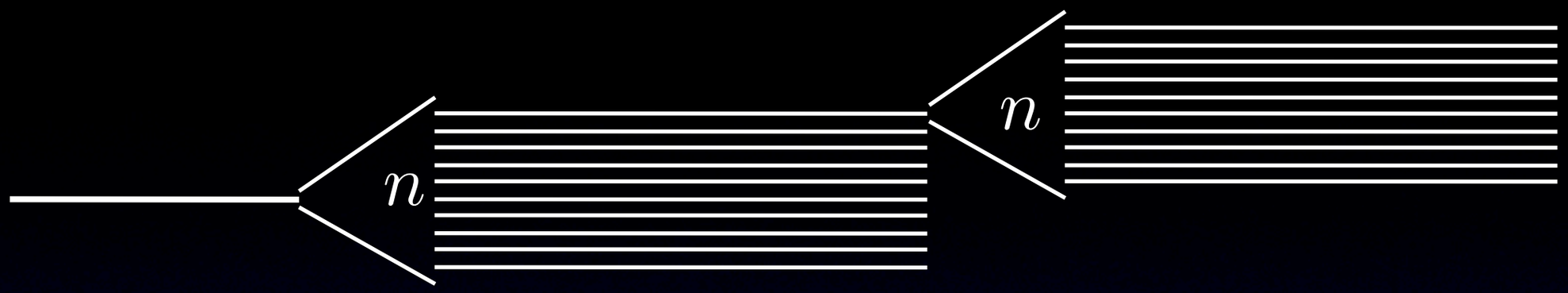
Perfection Through Concatenation



failure probability

p cp^2 $c(cp^2)^2$

Perfection Through Concatenation



failure probability

p $\quad cp^2 \quad c(cp^2)^2$

$$\frac{p_{\text{fail}}}{p_{\text{th}}} = \left(\frac{p}{p_{\text{th}}}\right)^{2^k} \qquad p_{\text{th}} = \frac{1}{c} \qquad N_{\text{qubits}} = n^k$$

$$N_{\text{qubits}} = \left[\log_{\frac{p}{p_{\text{th}}}} \left(\frac{p_{\text{th}}}{p_{\text{fail}}}\right)\right]^{\log_2 n} = O\left(\text{polylog } p_{\text{fail}}^{-1}\right)$$

Below threshold, failure suppressed exponentially as number of qubits

Can We Concatenate For a Self-Correcting Quantum Memory?

Stabilizer Codes in an Unstable Amount of Time

Stabilizer: group of commuting Pauli operators (not containing -I)

Ex: five qubit code

$$\mathcal{S} = \langle S_1, S_2, S_3, S_4 \rangle$$

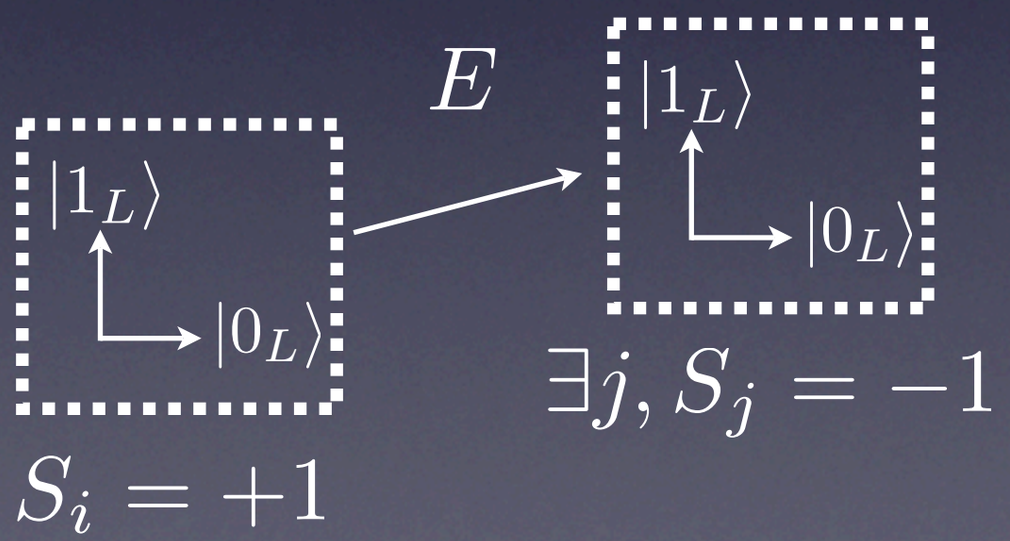
$$\begin{aligned} S_1 &= X \otimes Z \otimes Z \otimes X \otimes I \\ S_2 &= I \otimes X \otimes Z \otimes Z \otimes X \\ S_3 &= X \otimes I \otimes X \otimes Z \otimes Z \\ S_4 &= Z \otimes X \otimes I \otimes X \otimes Z \end{aligned}$$

Stabilizer code subspace:

$$S|\psi_C\rangle = |\psi_C\rangle, \ \forall S \in \mathcal{S}$$

Detectible errors:

$$\exists S \in \mathcal{S}, ES = -SE$$

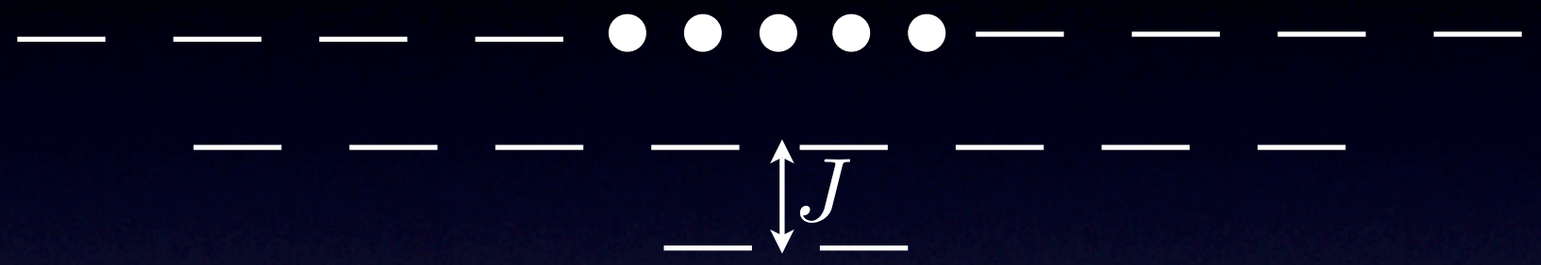


Stabilizer Hamiltonians

[See Kitaev, or more recently Jordan, Farhi, Shor]

$$H = -\frac{J}{2}\sum_{i=1}^{n-k} S_i$$

stabilizer generators



[[n,k,d]] code

➡ ground state is k fold degenerate

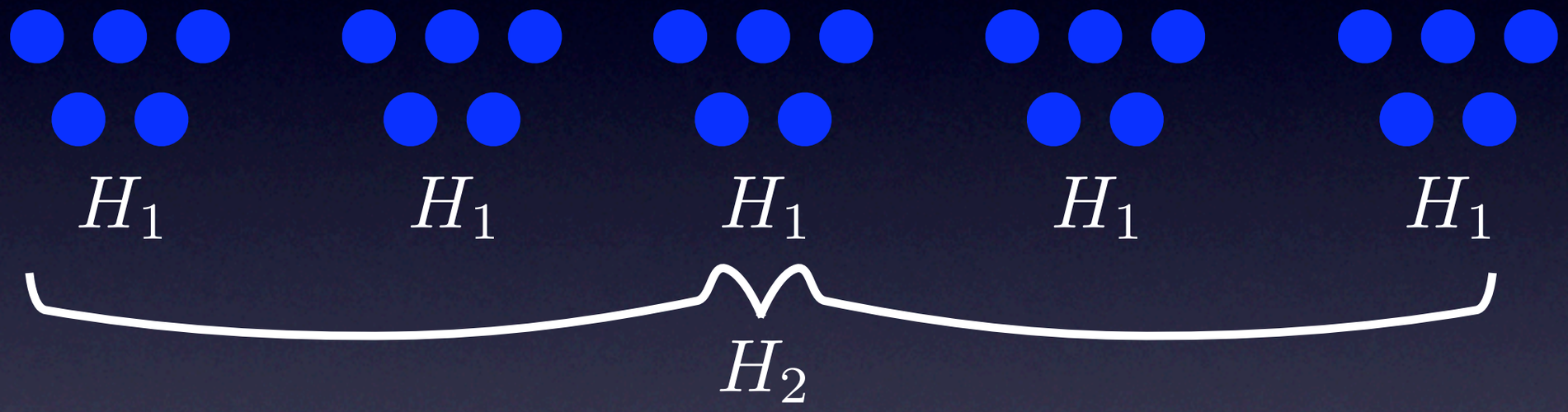
➡ smallest weight Pauli error which acts non-trivially on ground state subspace is weight k

Example:

$$\begin{aligned} H &= -\frac{J}{2}[X\otimes Z\otimes Z\otimes X\otimes I + I\otimes X\otimes Z\otimes Z\otimes X \\ &\quad + X\otimes I\otimes X\otimes Z\otimes Z + Z\otimes X\otimes I\otimes X\otimes Z] \end{aligned}$$

Concatenated Code Hamiltonians

[[n, 1, d]] base code

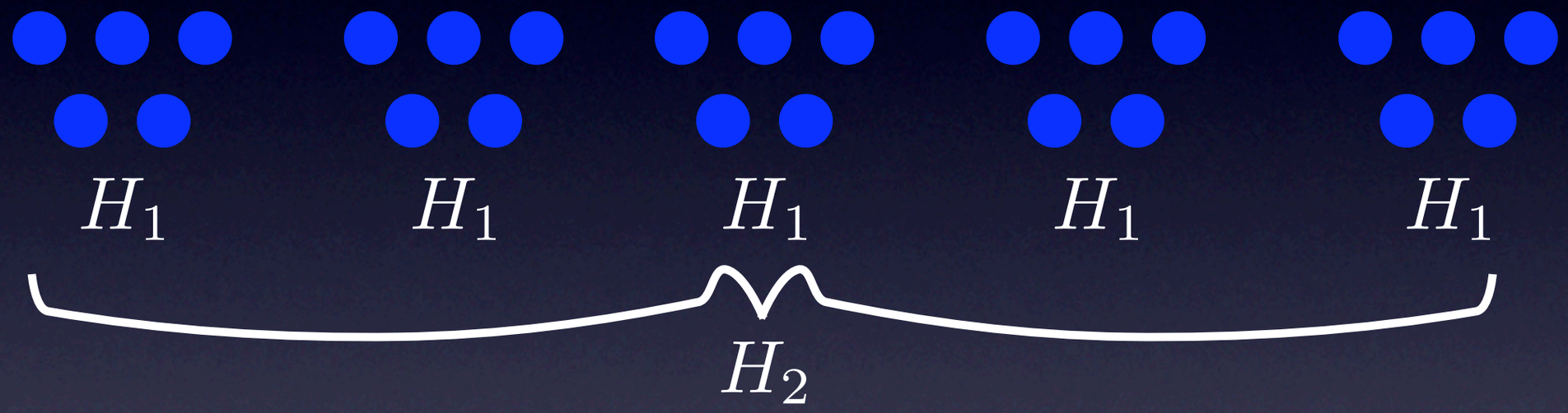


r levels of concatenation:

- n^r qubits
- Hamiltonian is sum of n^r-1 terms
- terms are up to weight c^r operators

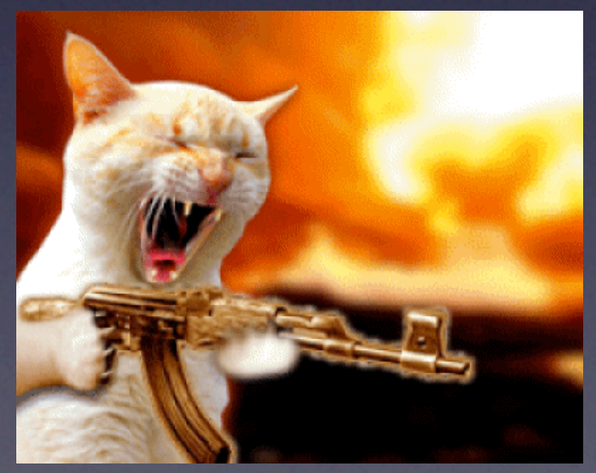
Concatenated Code Hamiltonians

[[n, 1, d]] base code



r levels of concatenation:

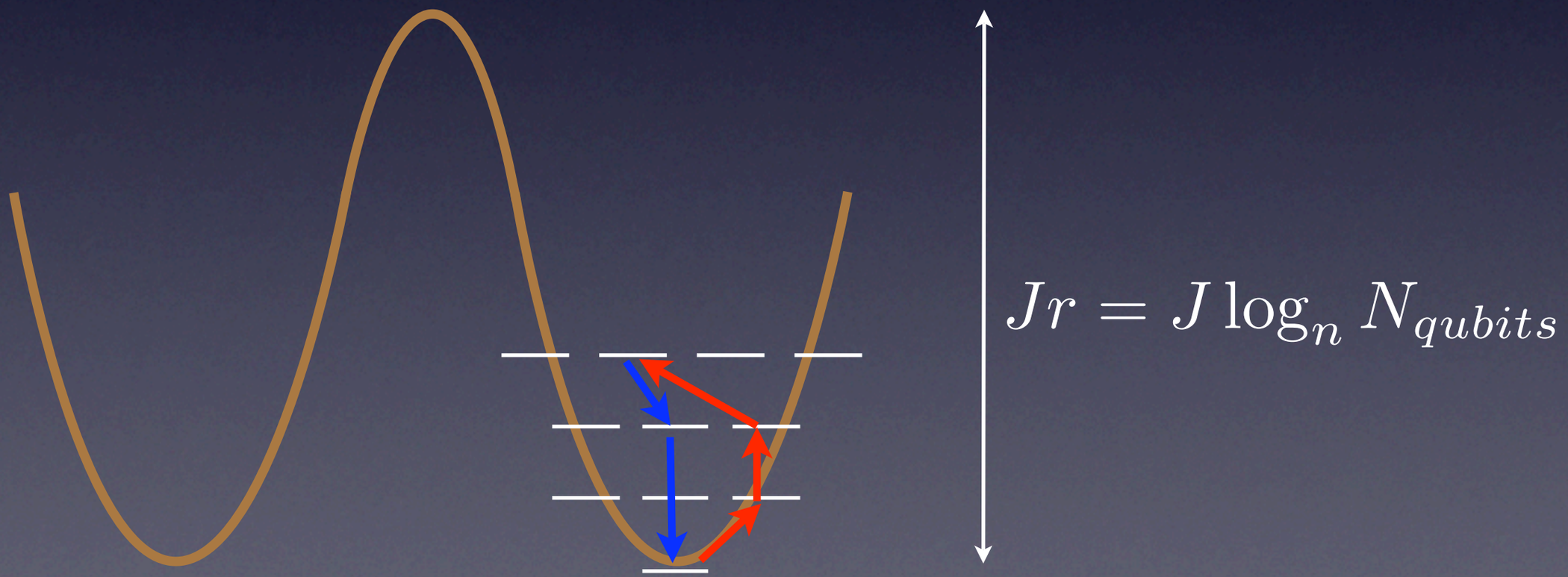
- n^r qubits
- Hamiltonian is sum of n^r-1 terms
- terms are up to weight c^r operators



Self-Correcting Barrier

Correction distance: number of errors corrected via the concatenated code correction scheme

Energy change bounded by correction distance.



Perturb It?



Want to show that local perturbations do not destroy information encoded into ground state.

$$H = H_{concat} + xV$$

But can we rigorously apply degenerate perturbation theory here?

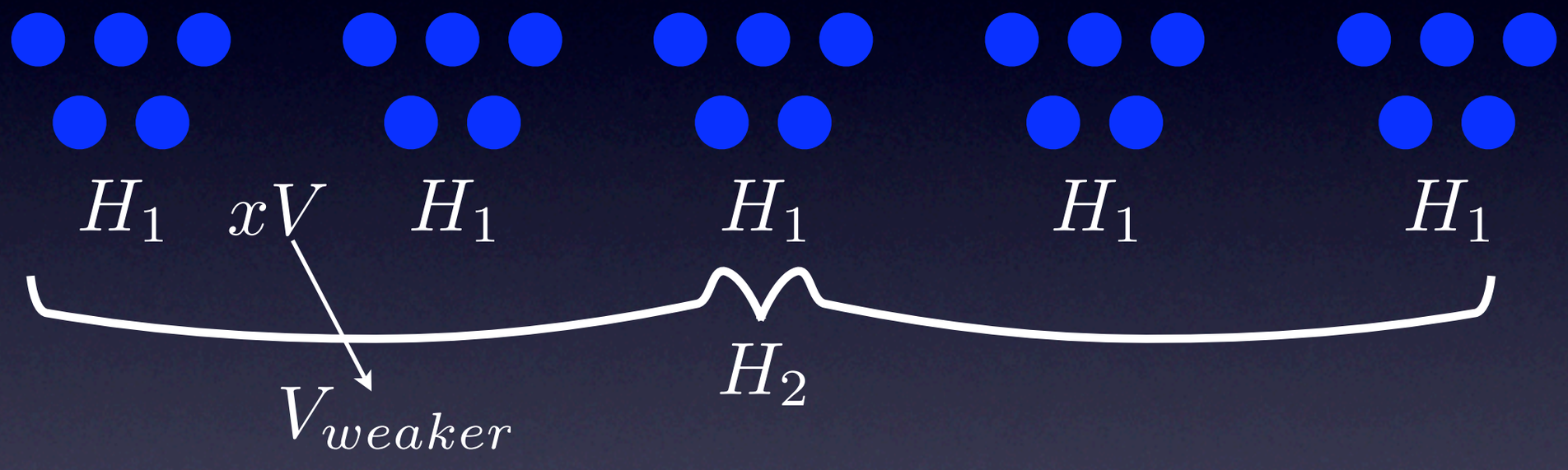
Kato/Bloch:

$||xV|| \ll \Delta$ perturbation which shrinks with system size

Schrieffer and Wolff:

perturbations on local Hamiltonians

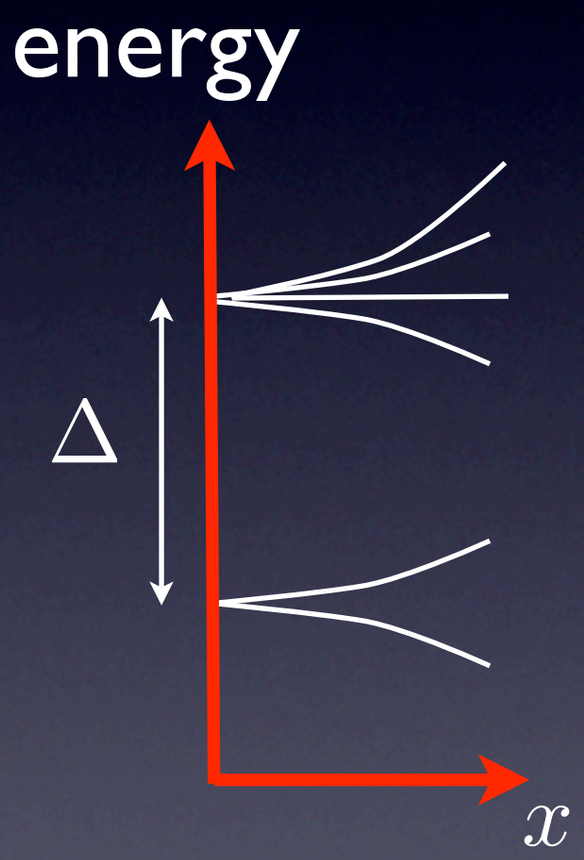
Leverage Concatenation





Deg. Pert. Theory

$$H = H_0 + xV \qquad H_0 = \sum_i E_i \Pi_i$$



If $||xV|| < \frac{\Delta}{2}$ then $H = \sum_i H_i^{\text{eff}}$

$$H_i^{\text{eff}} = \lambda_i P_i(x) + \sum_{m=1}^{\infty} x^m A_i^{(m)},$$

perturbed eigenspaces

convergent series

Deg. Pert. Theory

$$H_i^{\text{eff}} = \lambda_i P_i(x) + \sum_{m=1}^{\infty} x^m A_i^{(m)},$$

$$G_i^{(k)} = \begin{cases} \sum_{j \neq i} \frac{\Pi_j}{(\lambda_j - \lambda_i)^k} & \text{if } k > 0 \\ -\Pi_i & \text{if } k = 0 \end{cases}$$

$$\begin{aligned} A_i^{(1)} &= G_i^{(0)} V G_i^{(0)} \\ A_i^{(2)} &= -G_i^{(0)} V G_i^{(1)} V G_i^{(0)} - G_i^{(1)} V G_i^0 V G_i^{(0)} - G_i^{(0)} V G_i^{(0)} V G_i^{(1)} \end{aligned}$$

If we truncate we can bound the error:

$$\left\| \sum_{m=p+1}^{\infty} x^m A_i^{(m)} \right\| \leq \Delta \left(\frac{4x||V||}{\Delta} \right)^{p+1}$$

Ground State H_{eff}



Single qubit perturbations on [[n,1,3]] code:

$$xV = x\sum_{i=1}^{n}\sum_{Q\in\{X,Y,Z\}}\lambda_{Q,i}Q_i$$

$$\max_{Q,i}\lambda_{Q,i} = 1$$

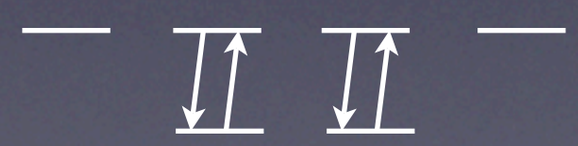
$$||V|| \leq 3n$$

First order:



$$xA_0^{(1)} = xG_0^{(0)}VG_0^{(0)} = 0$$

Second order:



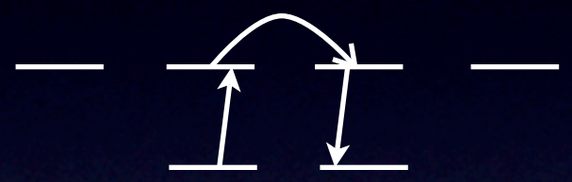
identity or gauge action

$$x^2A_0^{(2)} = x^2b_0\Pi_0$$

$$b_0 < \frac{\sigma_2}{J}$$

Ground State H_{eff}

Third order:



nontrivial term on g.s.

$$H_0^{\text{eff}} = E_0\Pi_0(x) + x^2 b_0\Pi_0 + M_0 \qquad ||M_0|| < J\left(\frac{12nx}{J}\right)^3 .$$

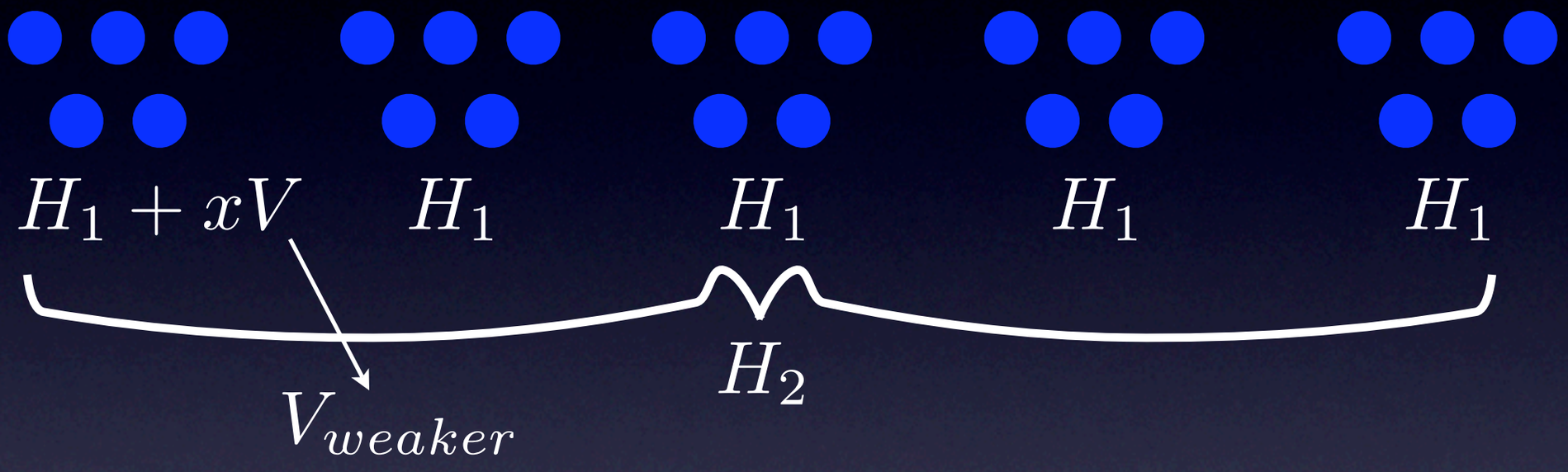
$$\sum_{L\in\{I,X,Y,Z\}} W_L \otimes L$$

where

$$||W_L|| < x^2 b_0 + J\left(\frac{12nx}{J}\right)^3$$

for x/J small enough....

But...



$$H_0 + xV = E_0 P_0(x) + x^2 b_0 \Pi_0 + M_0 + \sum_{i \neq 0} H_i^{\text{eff}}$$

$$\sum_{L \in \{I,X,Y,Z\}} W_L \otimes L$$

Higher States



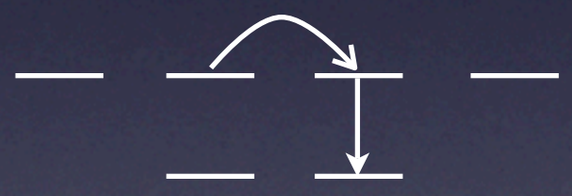
First order:



$$xA_i^{(1)} = xG_i^{(0)}VG_i^{(0)} = x\sum_{\alpha\neq\beta} a_i(\alpha,\beta)Q_{i,\alpha,\beta}$$

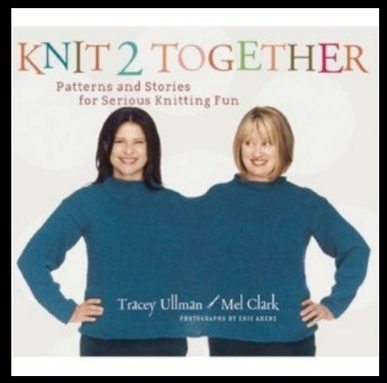
does not act on ground subspace

Second order:



$$\begin{aligned} x^2A_i^{(2)} &= x^2G_i^{(1)}VG_i^{(0)}VG_i^{(0)} + G_i^{(0)}VG_i^{(0)}VG_i^{(1)} \\ &= \sum_{i\neq j}\sum_{\alpha,\alpha'} x^2 b_{ij}(\alpha,\alpha')T_{i,j,\alpha,\alpha'}. \end{aligned}$$

Together



$$H = H'_{\text{stab}} + x^2 b_0 \Pi_0 + \sum_{i \neq 0} \left[x \sum_{\alpha \neq \beta} a_i(\alpha, \beta) Q_{i,\alpha,\beta} + x^2 \sum_{\alpha,\beta} b_i(\alpha,\beta) R_{i,\alpha,\beta} + x^2 \sum_{i \neq j} \sum_{\alpha,\alpha'} b_{ij}(\alpha,\alpha') T_{i,j,\alpha,\alpha'} \right] + M$$

focus on

$$\sum_{L \in \{I,X,Y,Z\}} W_L \otimes L$$

which act nontrivially on ground state

$$\sum_{L \in \{I,X,Y,Z\}} V_L \otimes L = x^2 b_0 \Pi_0 + x^2 \sum_{i \neq j} \sum_{\alpha,\alpha'} b_{ij}(\alpha,\alpha') T_{i,j,\alpha,\alpha'} + M$$

Threshold

$$||V_L|| \leq x^2|b_0| + x^2 \max_{i\neq 0} \max_{\alpha} |b_{0i}(0,\alpha')| + J\left(\frac{12nx}{J}\right)^3.$$

When is this lower strength than perturbation?

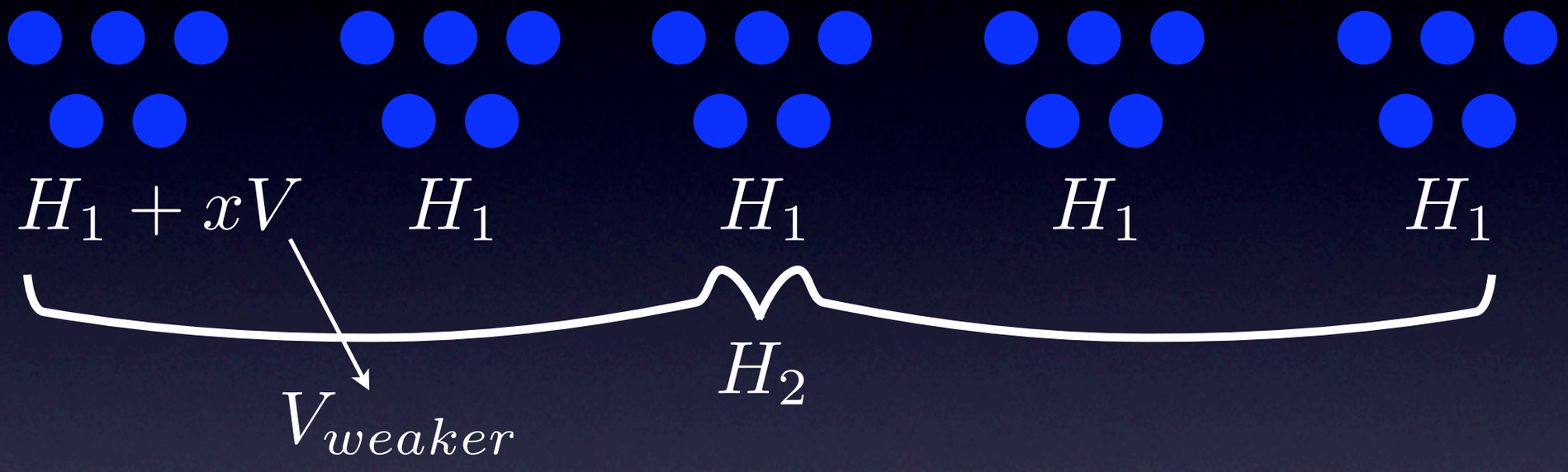
$$\|V_L\| < x$$

$$x^2|b_0| + x^2 \max_{i\neq 0} \max_{\alpha} |b_{0i}(0,\alpha')| + J\left(\frac{12nx}{J}\right)^3 < x$$

Combinatorial calculation for [[5,1,3]] code:

$$\frac{x}{J} < \frac{\sqrt{96081} - 9}{14400} \approx 0.0021$$

Review



r levels of concatenation: $x \to x^{2^r}$

Why?

- This seems to imply that concatenation produces a splitting at 2^r. Why isn't this d^r?
  - Probably technical: work with respect to unperturbed interaction
- What about the new effective stabilizers?
  - Similar analysis shows threshold in state being correctible

The Not So Hidden Problems



The Not So Hidden Problems

- n local interactions
- n local interactions
- n local interactions



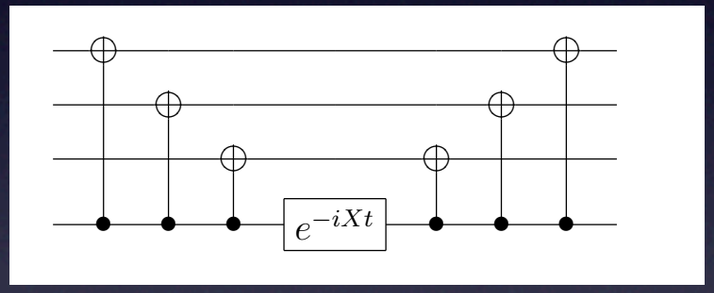
Solutions?

- Fault-tolerant simulation of stabilizer Hamiltonians
- Can we gadget our way out of this mess?

Simulate Hamiltonians

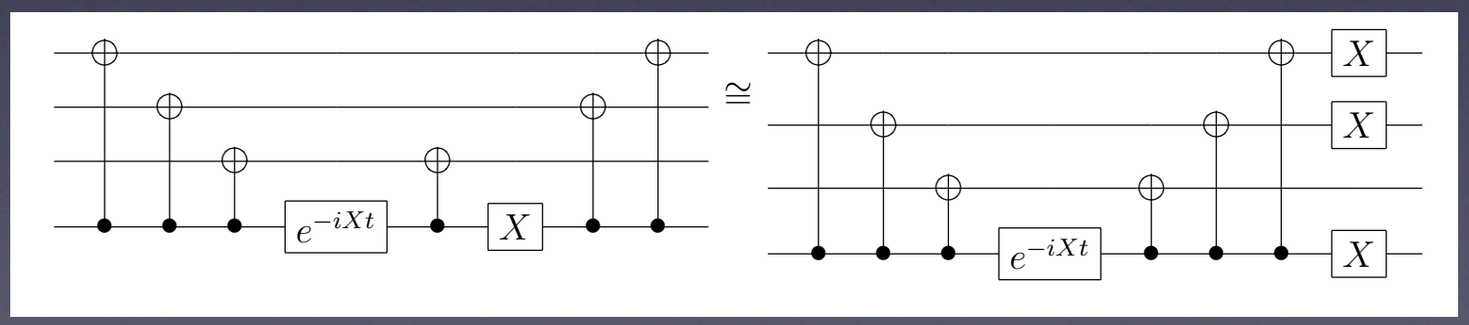
Lloyd, Rahn, Ahn, arXiv:quant-ph/9912040

Stroboscopic evolution to generate effective evolution



$$= \exp(-itX \otimes X \otimes X \otimes X)$$

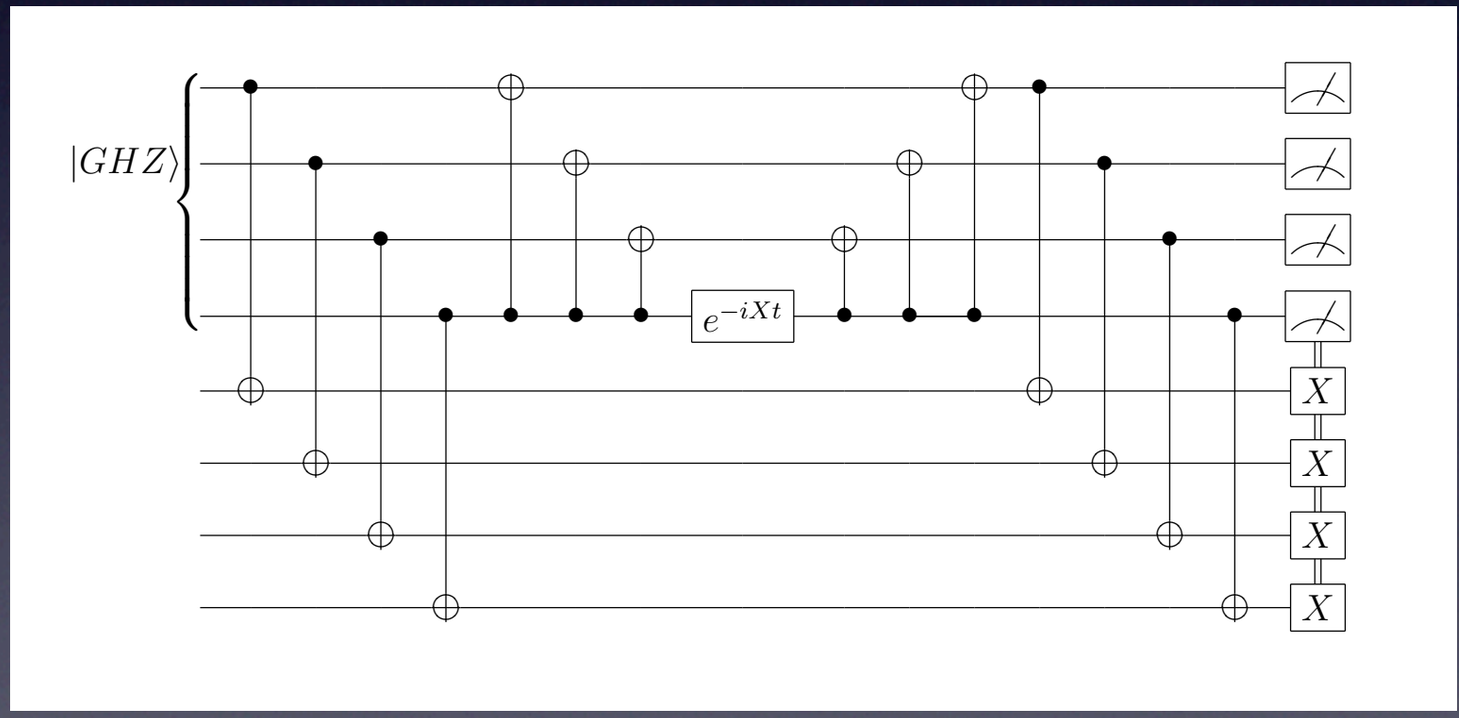
Problem:



single faults can spread to multiple faults

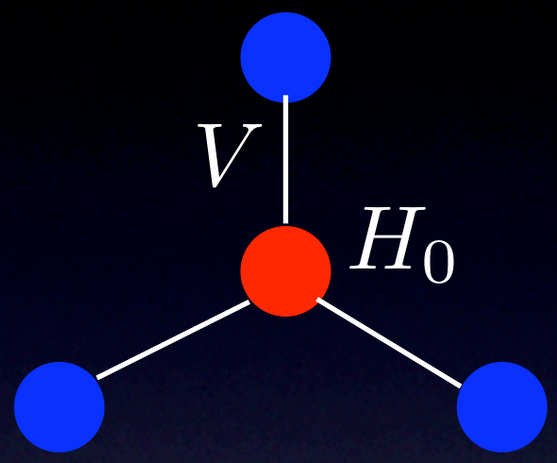
Fault-Tolerant Stabilizer Hamiltonian Simulation

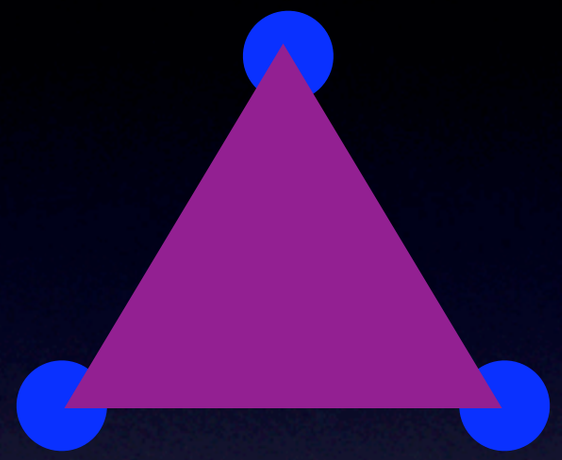
Similar conditions for fault-tolerant quantum error correction



Q: error model for this setup???

Gadgets

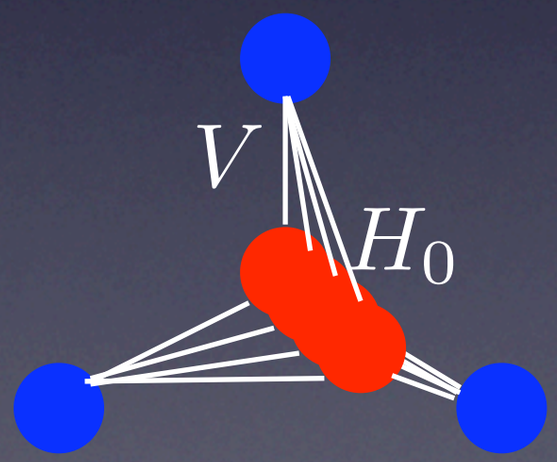




$$O\left(\frac{\lambda^3}{\Delta^2}\right)$$

three qubit interaction

Can we amplify?



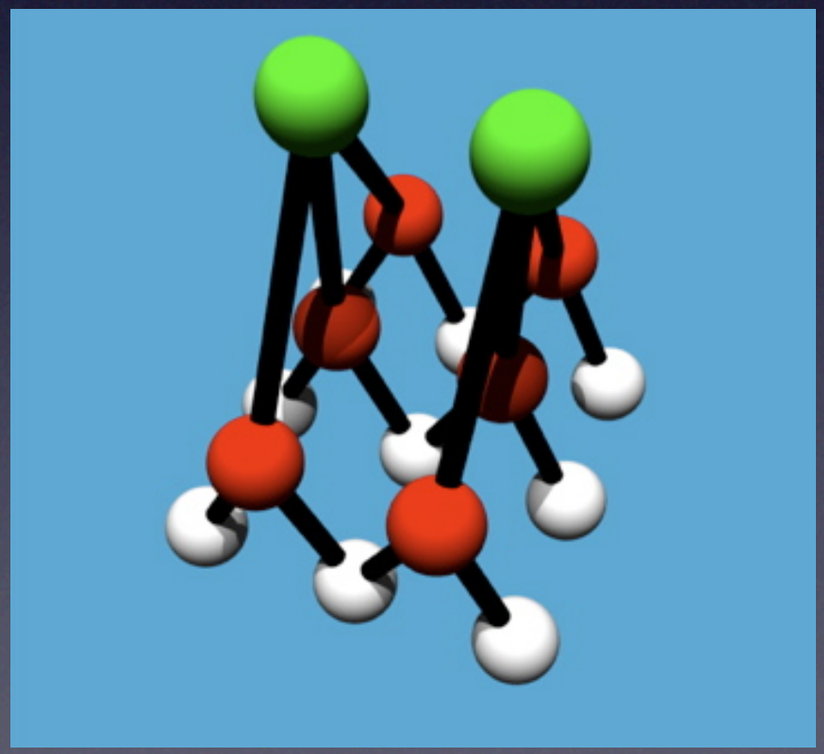
Note: important that individual gadgets commute

Single qubit errors on system "see" large bath

Concatenated Gadgets?

amplification simply replaces n qubit interactions with an exponential number of ancillas

Is it possible to get around this via concatenation?



A Final Observation

9 qubit subsystem code is well suited geometrically for concatenation

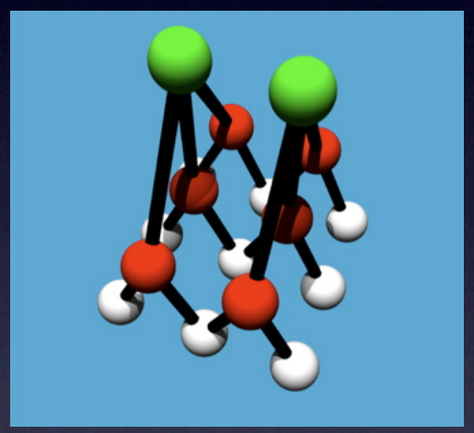
stabilizers

$$\begin{array}{ccc|ccc|ccc|ccc} X & X & X & I & I & I & Z & Z & I & I & Z & Z \\ X & X & X & X & X & X & Z & Z & I & I & Z & Z \\ I & I & I & X & X & X & Z & Z & I & I & Z & Z \end{array}$$

logical operators on boundary

$$\begin{array}{ccc|ccc} X & X & X & Z & I & I \\ I & I & I & Z & I & I \\ I & I & I & Z & I & I \end{array}$$

Crazy Quantum Concatenated Code Hamiltonians*



Dave Bacon
University of Washington

* This proposal, like all proposals for quantum computation, relies on speculative technology, does not in its current form take into account all possible sources of noise, unreliability and manufacturing error, and probably will not work.